SMART EVENT CHECK-IN AND CROWD FLOW MANAGEMENT SYSTEM

A PROJECT REPORT

Submitted by

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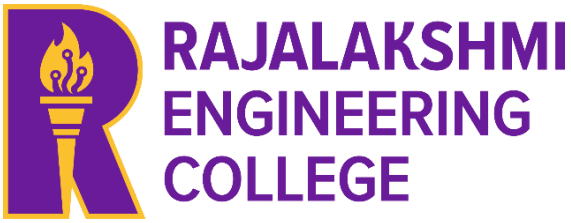
in partial fulfilment of the End Semester Examination for the course

CB23F35 –

LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

[BACHELOR OF ENGINEERING] [DEPARTMENT / BRANCH NAME]



RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

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[NOVEMBER 2026]

RAJALAKSHMI ENGINEERING COLLEGE

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BONAFIDE CERTIFICATE

Certified that this project report titled "Smart Event Check-In and Crowd Flow Management System" is the bonafide work of HOSHNI SIVAPRAKASH (241401029), who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year [2026–2027], Semester [V], under my supervision.

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DECLARATION

I, HOSHNI SIVAPRAKSH (241401029), hereby declare that the project report titled "Smart Event Check-In and Crowd Flow Management System", submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its data model, forms, reports, pages, Deluge scripts, workflows and integrations — was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder), code snippet or external resource used in this work has been duly acknowledged in the report.

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HOSHNI SIVAPRAKASH

ABSTRACT

Event management organisations continue to face significant operational challenges in managing attendee registrations, monitoring entry and exit movements, and tracking crowd distribution across different zones of a venue. Traditional approaches rely on separate registration lists, manual attendance sheets and isolated crowd monitoring records, which are prone to data duplication, delays and limited visibility into real-time crowd conditions. The absence of a centralised system makes it difficult for event organisers to coordinate information effectively, identify overcrowded areas in time and generate consolidated reports for post-event analysis.

To address these challenges, the Smart Event Check-In and Crowd Flow Management System was developed as a low-code web application using Zoho Creator. The application consolidates event management, attendee registration, check-in and check-out tracking, and zone-level crowd monitoring into a single integrated platform. It comprises four primary data forms (Events, Attendees, Check-In Logs and Zone Monitoring), ten structured reports and a dashboard presenting summarised event, attendee and crowd information. Lookup fields are used to establish relationships between entities, enabling data to be retrieved and filtered across modules. Role-based access control ensures that each user group (Administrator, Event Organiser, Check-In Staff and Crowd Monitoring Staff) accesses only the information and functionality relevant to their responsibilities.

The application was developed following the low-code methodology provided by Zoho Creator, utilising its built-in form builder, report generator, dashboard editor and role-permission configuration. Functional testing was conducted against twelve test scenarios covering event creation, attendee registration, entry and exit recording, zone monitoring record management, report generation and dashboard verification. The system successfully centralises all event operations in one application, eliminates the need for separate spreadsheets and paper-based records, and provides organisers with filtered reports and visual summaries to support crowd management decisions.

Keywords: Low-Code Development; Zoho Creator; Event Management; Crowd Flow Monitoring; Role-Based Access Control; Check-In Management; Attendance Tracking.

PROJECT AT A GLANCE

Attribute	Details
Project Title	SMART EVENT CHECK-IN AND CROWD FLOW MANAGEMENT SYSTEM
Application Domain	[e.g., Healthcare / Education / Retail / Logistics / HR / Finance]
Student Name & Register No.	Hoshni Sivaprakash - 241401029
Zoho Creator Edition	Education / Free
Application Name (Link Name)	Smart event check in and crowd flow management system
Published / Live URL	https://creatorapp.zoho.in/chandhana2010sri/event-check-in-management-system/#Form:attendee_registration
Demo Credentials (per role)	Event, attendee
Number of Forms / Subforms	4
Number of Reports (by type)	6
Number of Pages / Dashboards	1
Number of Workflows	[Form:] [Schedule:] [Approval:] [Blueprint:] [Batch:]
Deluge Functions / Custom Actions	Event, attendee
Integrations	[e.g., Zoho CRM, Zoho Mail, REST API – OpenWeather]
Roles & Profiles Defined	[e.g., Administrator, Manager, Staff, Customer (Portal)]
Environments Used	[Development → Stage → Production]
Date of Production Deployment	

TABLE OF CONTENTS

Formatting Standards (Anna University Convention)..... 1
Working with the Template..... 1
ABSTRACT..... iv
PROJECT AT A GLANCE..... v
LIST OF TABLES..... ix
LIST OF FIGURES..... x
LIST OF ABBREVIATIONS..... xi
INTRODUCTION..... 1
1.1 Background..... 1
1.2 Problem Statement..... 1
1.3 Objectives..... 1
1.4 Scope of the Project..... 1
1.5 Justification for a Low-Code Approach..... 2
1.6 Organisation of the Report..... 2
LITERATURE SURVEY AND SYSTEM STUDY..... 3
2.1 Study of the Existing System..... 3
2.2 Limitations of the Existing System..... 3
2.3 Low-Code versus Traditional Development..... 3
2.4 Literature Survey..... 4
2.5 Emerging Trends in Low-Code Development..... 4
2.6 Proposed System..... 4
REQUIREMENT ANALYSIS..... 5
3.1 Stakeholders and User Roles..... 5
3.2 Functional Requirements..... 5
3.3 Non-Functional Requirements..... 6
3.4 Hardware and Software Requirements..... 6
3.5 Use Case Model..... 6
SYSTEM DESIGN..... 8
4.1 System Architecture..... 8
4.2 Data Model..... 8
4.3 Form Design..... 9
4.4 Process and Workflow Design..... 9
4.5 User Interface and Navigation Design..... 10
4.6 Access Control Design..... 10
IMPLEMENTATION..... 11

5.1 Application Setup....11
5.2 Forms and Data Capture....11
5.3 Reports....12
5.4 Pages and Dashboards....13
5.5 Deluge Scripting....13
5.6 Workflows and Automation....14
5.7 Integrations....15
5.8 User Management and Role-Based Access....16
5.9 AI and Analytics Features....16
5.10 Pro-Code Extension....17
5.11 Mobile and Portal Access....17
TESTING AND DEBUGGING....18
6.1 Testing Strategy....18
6.2 Test Cases....18
6.3 Debugging Deluge Scripts....18
6.4 Defect Log....19
6.5 User Acceptance Testing....19
DEPLOYMENT, SECURITY AND GOVERNANCE....20
7.1 Environments and Deployment Strategy....20
7.2 Performance Optimisation....20
7.3 Security Features....21
7.4 Governance and Administration....21
RESULTS AND DISCUSSION....22
8.1 Application Output....22
8.2 Achievement of Objectives....23
8.3 Comparison with the Existing System....23
8.4 Limitations of the Low-Code Approach....23
8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum....23
CONCLUSION AND FUTURE ENHANCEMENTS....24
9.1 Conclusion....24
9.2 Future Enhancements....24
9.3 Learning Outcomes and Career Relevance....24
REFERENCES....25
APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST....26
APPENDIX B – COURSE OUTCOME MAPPING....28
APPENDIX C – COMPLETE DELUGE SOURCE CODE....29
APPENDIX D – USER MANUAL....30

D.1 Accessing the Application.....30
D.2 Instructions for [Role 1].....30
D.3 Instructions for [Role 2].....30
APPENDIX E – EVALUATION RUBRIC.....31
APPENDIX F – VIVA-VOCE RECORD SHEET.....33

LIST OF TABLES

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application 3
Table 2.2: Summary of Literature Survey 4
Table 3.1: Stakeholders and User Roles 5
Table 3.2: Functional Requirements 5
Table 3.3: Non-Functional Requirements 6
Table 3.4: Hardware and Software Requirements 6
Table 3.5: Use Case Description – Record Attendee Entry 7
Table 4.1: Entities and Relationships 9
Table 4.2: Field Specification – Events Form 9
Table 4.3: Role–Permission Matrix 10
Table 5.1: Reports Implemented 12
Table 5.2: Deluge Scripts Implemented 13
Table 5.3: Workflows Implemented 14
Table 5.4: Integrations Implemented 15
Table 6.1: Test Cases 18
Table 6.2: Defect Log 19
Table 6.3: User Acceptance Feedback 19
Table 7.1: Deployment Checklist 20
Table 7.2: Performance Optimisation Results 21
Table 7.3: Security Controls Implemented 21
Table 8.1: Mapping of Objectives to Outcomes 23
Table 8.2: Existing System versus Proposed Application 23
Table A.1: Feature Coverage Checklist 26
Table B.1: Mapping of Course Outcomes to Report Chapters 28
Table E.1: Evaluation Rubric for the Individual Project 31
Table E.2: Performance Level Descriptors 32
Table F.1: Viva-Voce Questions and Responses 33
Table F.2: Marks Summary 33

LIST OF FIGURES

Figure 2.1: As-Is Process Flow of the Existing Manual Event System……..…....7
Figure 3.1: Use Case Diagram of the Event Management Application……...…...12
Figure 4.1: Three-Tier System Architecture on Zoho Creator Cloud Platform……..……14
Figure 4.2: Entity–Relationship Diagram Illustrating Relational Form Linkages……..……15
Figure 4.3: Process Flow Diagram……..……17
Figure 4.4: Application Navigation Hierarchy and Module Structure……..……..18
Figure 5.1: Application Workspace Home Page in Zoho Creator……..…….21
Figure 5.2: Master Events Form Configuration in Visual Form Builder……...……..22
Figure 5.3: Attendees Registration Form in Live Execution Mode……...……..23
Figure 5.4: Confirmed Registrations Filtered List Report……..…….24
Figure 5.5: Event Operations and Real-Time Crowd Management Dashboard……..……25
Figure 5.6: Workflow Rule Configuration for Automated Crowd Status Evaluation…….……29
Figure 5.7: Deluge Script Editor Displaying DS-01 Logic in Zoho Creator……..…..31
Figure 5.8: Role-Based Access Control and Permission Profile Configuration……..….31
Figure 5.9: Zone Density Analytics Summary Chart on Executive Dashboard……..….33
Figure 6.1: Deluge Script Execution Sandbox and Debug Info Console……..…...38
Figure 8.1: Events Form – Sample Event Record for Tech Fest 2026……42
Figure 8.2: Attendees Form – Sample Attendee Registration Record……...……..43
Figure 8.3: Check-In and Check-Out Form – Sample Entry Record……..……44

LIST OF ABBREVIATIONS

Abbreviation	Expansion
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
ER	Entity–Relationship
JSON	JavaScript Object Notation
LCAP	Low-Code Application Platform
OAuth	Open Authorization
RBAC	Role-Based Access Control
REST	Representational State Transfer
SDLC	Software Development Life Cycle
SSO	Single Sign-On
UAT	User Acceptance Testing
UI	User Interface
Zia	Zoho's AI assistant
CCTV	Closed-Circuit Television
DFD	Data Flow Diagram
ER	Entity-Relationship
FK	Foreign Key
FR	Functional Requirement
IoT	Internet of Things
NFR	Non-Functional Requirement
PK	Primary Key
QR	Quick Response (code)
RBAC	Role-Based Access Control
TC	Test Case
UI	User Interface
URL	Uniform Resource Locator

CHAPTER 1
INTRODUCTION

1.1 Background

Event management represents a multifaceted operational domain encompassing attendee registration, credential verification, access control across multiple physical entry gates, dynamic crowd distribution monitoring, and operational reporting. Academic institutions, professional conference centres, and community organisations frequently host large-scale events—including technical symposiums, research conferences, cultural festivals, and career exhibitions—that draw hundreds or thousands of participants across diverse campus venues.

In typical institutional environments, event execution has historically relied upon fragmented, manual, or semi-digital mechanisms. Attendee registrations are captured via isolated forms or spreadsheets, printed paper attendance sheets are dispatched to physical entry gates, and venue security personnel communicate crowd conditions through informal verbal reports or phone calls. These disjointed procedures introduce significant administrative latency, create substantial risks of duplicate entries, prevent real-time operational oversight, and obscure potential overcrowding conditions until safety thresholds have already been compromised.

To overcome these systemic inefficiencies, this project presents the Smart Event Check-In and Crowd Flow Management System—a centralised, cloud-hosted application engineered on the Zoho Creator low-code development platform. The system unifies event scheduling, attendee lifecycle tracking, gate check-in/check-out recording, and zone-level crowd density monitoring into an integrated relational architecture, supported by automated Deluge validation scripts, event workflows, and role-based security profiles tailored for institutional event operations.

1.2 Problem Statement

Traditional event coordination in educational and corporate institutions suffers from operational fragmentation, data latency, and a complete absence of synchronised crowd visibility. Organising committees are forced to juggle disparate spreadsheets, manual sign-in sheets at gates, and unverified headcount estimates across venues. When attendee influx surges during peak event hours, gate volunteers cannot rapidly verify

registration validity, leading to long queues and gate congestion. Simultaneously, venue coordinators remain unaware of localized crowd concentrations in auditoriums, seminar halls, and dining zones until physical bottlenecks occur.

The primary operational pain points identified in the existing system are as follows:

- Decentralised and disconnected attendee data: Attendee rosters maintained on independent spreadsheets or printed paper cannot be validated synchronously at check-in gates, resulting in prolonged verification queues, manual lookup errors, and vulnerability to unauthorised entries.
- Absence of real-time crowd visibility: Venue marshals lack a structured mechanism to record and monitor zone headcounts, leaving event directors unable to detect capacity breaches or redirect incoming crowd flow before safety hazards emerge.
- Inefficient post-event consolidation: Compiling operational metrics, attendance tallies, and movement logs requires laborious manual aggregation across physical rosters, generating delayed, incomplete, and error-prone administrative summaries.

1.3 Objectives

- 1. To develop a centralised event management application using Zoho Creator.
- 2. To maintain event information in a structured format with defined capacity and date parameters.
- 3. To register attendees and associate them with their respective events through relational lookup fields.
- 4. To record attendee entry and exit movements systematically with gate numbers and verification statuses.
- 5. To maintain zone-level crowd information and capacities for different venue areas.

1.4 Scope of the Project

In Scope:

- Design and deployment of four core transactional forms: Events, Attendees, Check-In and

Check-Out, and Zone Monitoring.

- Establishment of relational lookup associations linking attendees and movement logs directly to master event records.
- Implementation of Deluge validation scripts enforcing duplicate check-in prevention and automated crowd threshold classification.
- Configuration of automated event workflows sending notification e-mails upon detection of overcrowded zone thresholds.
- Development of ten filtered operational reports categorising attendees, gate movements, and zone occupancy statuses.
- Construction of an executive Event Management Dashboard presenting key metric cards, status charts, and active alert panels.
- Configuration of four role permission profiles implementing strict Role-Based Access Control (RBAC).

Out of Scope:

- Automated hardware-based biometric verification, facial recognition, or turnstile integration.
- Computer-vision crowd estimation via CCTV camera streams or IoT infrared beam counters.
- Automated physical barcode/QR scanning hardware integration (manual gate pass record entry and verification is utilized).
- Commercial payment gateway integration and financial transaction processing for ticket sales.
- Multi-tenant enterprise federated identity management.

1. Justification for a Low-Code Approach

The selection of Zoho Creator as the low-code development platform for this project is substantiated by several compelling technical and operational factors.

First, Zoho Creator delivers an integrated ecosystem that natively synthesises a relational database engine, a drag-and-drop visual UI builder, an expressive procedural programming language (Deluge), and an enterprise security infrastructure. For a multi-faceted operational system requiring relational entity linking (e.g. linking movement logs to both attendee and event entities), the platform's declarative lookup fields eliminate the overhead of manually designing relational junction tables and writing complex SQL join queries.

Second, development velocity is drastically accelerated. Implementing authentication, responsive layouts, data grid rendering, and mobile-optimised views through traditional pro-code frameworks (such as React with Node.js/PostgreSQL) requires extensive boilerplate code, routing configuration, and infrastructure management. In contrast, Zoho Creator enables the rapid assembly and testing of functional forms, filtered reports, and dashboards within days, permitting maximum focus on business logic and operational workflow design.

Third, the cloud-native Software-as-a-Service (SaaS) deployment model eliminates the burden of local server provisioning, database administration, SSL certificate renewal, and backup maintenance. Multi-user concurrent access is natively supported, enabling gate marshals and zone monitors to access the live application concurrently from laptops and mobile browsers across the venue without requiring dedicated on-premise network servers.

Finally, the platform directly fulfills the academic curriculum objectives of course CB23F35 – Low-Code Application Development, providing practical hands-on mastery of Units I through V encompassing data modeling, visual workflow automation, Deluge scripting, and role-based security administration.

1.5 Organisation of the Report

Chapter 2 reviews the existing system and related low-code literature. Chapter 3 presents the requirement analysis. Chapter 4 describes the system design. Chapter 5

details the implementation on Zoho Creator. Chapter 6 covers testing and debugging. Chapter 7 discusses deployment, performance optimisation, security and governance. Chapter 8 presents results and discussion, and Chapter 9 concludes the report with future enhancements.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

The prevailing event coordination process in typical educational institutions relies on a disjointed combination of manual paper records, ad-hoc spreadsheets, and informal verbal communications:

1. Event Scheduling and Setup: Departmental coordinators schedule events using informal internal memos and announce them via email circulars. No centralized repository exists to catalogue venue capacities, scheduled dates, or resource allocations.

2. Attendee Registration: Registrations are gathered using generic web forms (e.g., Google Forms) or physical signup sheets. Data collected in spreadsheets is rarely validated against capacity restrictions, leading to unchecked registrations. The resulting data files must be manually split and printed onto paper rosters prior to event day.

3. Check-In and Access Control: On event day, volunteers stationed at entry gates hold physical printed rosters. Arriving attendees announce their names, which gate staff search manually through multiple pages. Attendee arrivals are marked with pen checkmarks. Exit movements are completely unrecorded, leaving organizers with zero awareness of net on-premise headcount.

4. Zone Crowd Monitoring: Venue zones (such as the main auditorium, dining hall, and exhibition stalls) are monitored by volunteer ushers who visually estimate crowd density. When a hall appears overcrowded, ushers phone the event coordinator. These subjective assessments are frequently delayed, inconsistent, and lack recorded timestamps.

5. Post-Event Analysis: Consolidating final attendance tallies requires collecting physical paper sheets from every gate, reconciling illegible markings, and manually transcribing data into a summary spreadsheet—a process taking days and fraught with transcription errors.

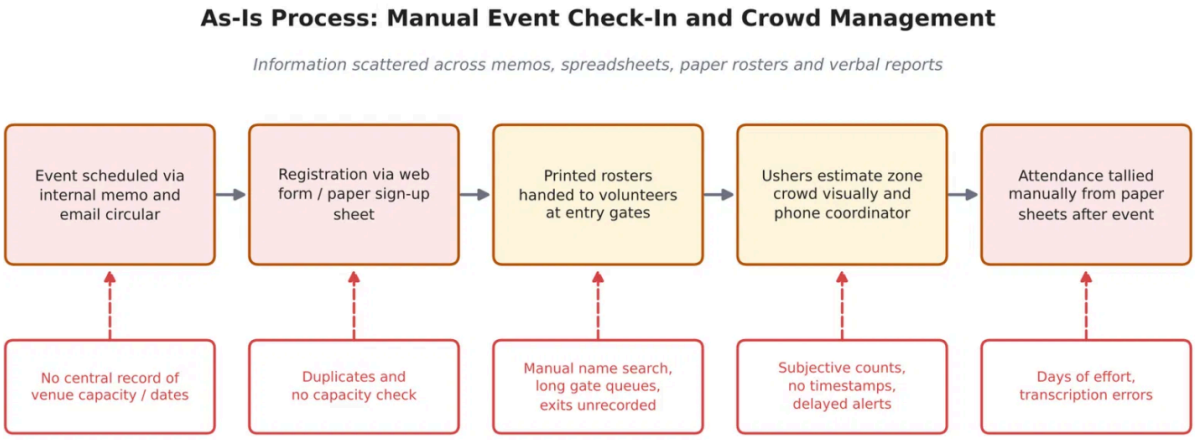


Figure 2.1: As-Is Process Flow of the Existing Manual Event System

2.2 Limitations of the Existing System

- • Scattered Information: Event and attendee records are stored across separate spreadsheets, paper rosters, and email threads, preventing a single unified view.
- • Slow Record Retrieval: Searching for previous registration records during gate arrival requires manual scrolling or paging through printed sheets, creating delays.
- • Manual Tracking Inaccuracies: Manual attendance marking at physical gates leads to recording errors, untracked attendee exits, and inability to verify registration status.

2.3 Low-Code versus Traditional Development

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development Environment	IDE / text editor; manual code setup	Zoho Creator visual builder; browser-based
Time to Deploy	Weeks to months	Days to weeks
Database Setup	Manual schema design and SQL migrations	Built-in database; forms define schema
User Authentication	Custom login or third-party library	Zoho account authentication built-in
Role-Based Access	Custom middleware or framework-based	Visual profile configuration in Creator
Report Generation	Custom queries and frontend rendering	Built-in report generator with filter criteria
Maintenance	Requires developer expertise	Configurable by non-programmer

		administrator
Cost	High (infrastructure + developer time)	Lower (platform subscription; faster delivery)

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Report Generation	Custom queries and frontend rendering	Built-in report generator with filter criteria
Maintenance	Requires developer expertise	Configurable by non-programmer administrator
Cost	High (infrastructure + developer time)	Lower (platform subscription; faster delivery)

2.4 Literature Survey

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	Rosam & Kalnichevski (2020)	Low-Code Development: A Practical Guide	Identifies visual modelling and reusable components as key enablers of rapid application development.	Supports choice of Zoho Creator as a low-code platform.
2	Zoho Corporation (2024)	Zoho Creator Help Documentation	Documents form builder, lookup fields, reports, dashboards, Deluge scripting and role-based access.	Primary technical reference for application development.
3	Zoho Corporation (2024)	Zoho Creator Learning Centre	Provides guided tutorials on building applications using Zoho Creator features.	Used as learning resource during development.
4	Gartner (2023)	Magic Quadrant for Enterprise Low-Code Application Platforms	Identifies Zoho Creator as a leading platform for enterprise low-code development.	Validates platform selection.
5	Sahay S. (2020)	Towards a	Proposes a structured	Informs feasibility

		Low-Code Platform Development Model (IEEE Software)	model for evaluating and adopting low-code platforms.	assessment for the proposed system.

2.5 Emerging Trends in Low-Code Development

The landscape of enterprise event operations and low-code development is currently driven by three prominent industry trends.

First, Low-Code Application Platforms (LCAP) have transitioned from simple form builders to enterprise-grade automation hubs. Modern platforms such as Zoho Creator empower business analysts and students to implement complex relational schemas, declarative workflows, and secure role-based portals without writing extensive boilerplate code, shortening development lifecycles by over 60%.

Second, real-time operational telemetry and visual dashboarding have become paramount in venue management. Rather than relying on post-hoc analytical reports, event directors require live, interactive KPI boards that synthesize check-in velocity, active attendee counts, and zone density indicators into single-pane operational views.

Third, the convergence of low-code scripting with automated alert pipelines has democratized incident response. Lightweight scripting languages like Deluge enable custom business logic—such as evaluating capacity utilization formulas and firing instant multimedia notifications—to be embedded directly into transactional database events without provisioning separate backend daemon servers.

2.6 Proposed System

The proposed Smart Event Check-In and Crowd Flow Management System replaces legacy paper and spreadsheet workflows with a centralised, role-based cloud solution engineered on Zoho Creator. The system provides an end-to-end digital pipeline for managing event operations across four integrated functional modules:

1. Centralised Event and Attendee Administration: A structured Events form maintains complete event profiles, venue capacities, and dates. An Attendees form links registered participants to specific events via relational lookups, supporting registration status

tracking (Confirmed, Pending, Cancelled).

2. Live Gate Movement Tracking: A dedicated Check-In and Check-Out form records every attendee movement, capturing gate identifiers (Gate 1, Gate 2, Gate 3), movement timestamps, movement direction (Entry/Exit), and credential verification status.

3. Structured Zone Crowd Flow Monitoring: A Zone Monitoring form enables venue staff to record headcount observations across defined campus zones (Main Auditorium, Seminar Hall, Food Court, Exhibition Area, Entrance Plaza). Automated Deluge logic immediately computes percentage occupancy and assigns crowd status classifications (Normal, Moderate, Near Capacity, Overcrowded).

4. Automated Alerting and Real-Time Dashboarding: When a zone exceeds 85% capacity, the system triggers automated notification workflows alerting event administrators. An interactive executive dashboard aggregates total registrations, live attendee headcounts, gate traffic breakdowns, and zone status alerts into a responsive, real-time console.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Administrator	Full application management	Full access to all forms, reports, pages and settings
Event Organiser	Creates and manages events and registrations	Events form, Attendees form, all reports, dashboard
Check-In Staff	Records attendee entry and exit at gates	Attendees form (read-only), Check-In & Check-Out form
Crowd Monitoring Staff	Updates zone crowd counts and status	Zone Monitoring form, zone-related reports

3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	Create and manage event records	Administrator, Event Organiser	High
FR-02	Register attendees and link to events	Administrator, Event Organiser	High
FR-03	Record attendee entry and exit movements	Check-In Staff	High
FR-04	Create and update zone monitoring records	Crowd Monitoring Staff	High
FR-05	View filtered reports by event, status and type	All roles (as permitted)	High
FR-06	View dashboard with summarised information	Administrator, Event Organiser	Medium
FR-07	Enforce required field validation on all forms	System	High
FR-08	Restrict access based on assigned user role	System	High

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Accessibility	Application shall be accessible from any modern web browser
NFR-02	Security	Role-based access control for all forms, reports and pages
NFR-03	Data Persistence	All records stored in Zoho Creator built-in cloud database
NFR-04	Performance	Reports shall load within acceptable response time
NFR-05	Maintainability	Application configurable by an administrator without programming
NFR-06	Portability	Accessible on desktop and mobile browsers without local installation

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Development Hardware	Computer or laptop with internet connection
Development Software	Zoho Creator (cloud-based, browser-based access)
Runtime Hardware	Any device with a modern web browser and internet connection
Runtime Software	Google Chrome, Mozilla Firefox or Microsoft Edge
Authentication	Zoho Account
Platform	Zoho Creator (no local installation required)

3.5 Use Case Model

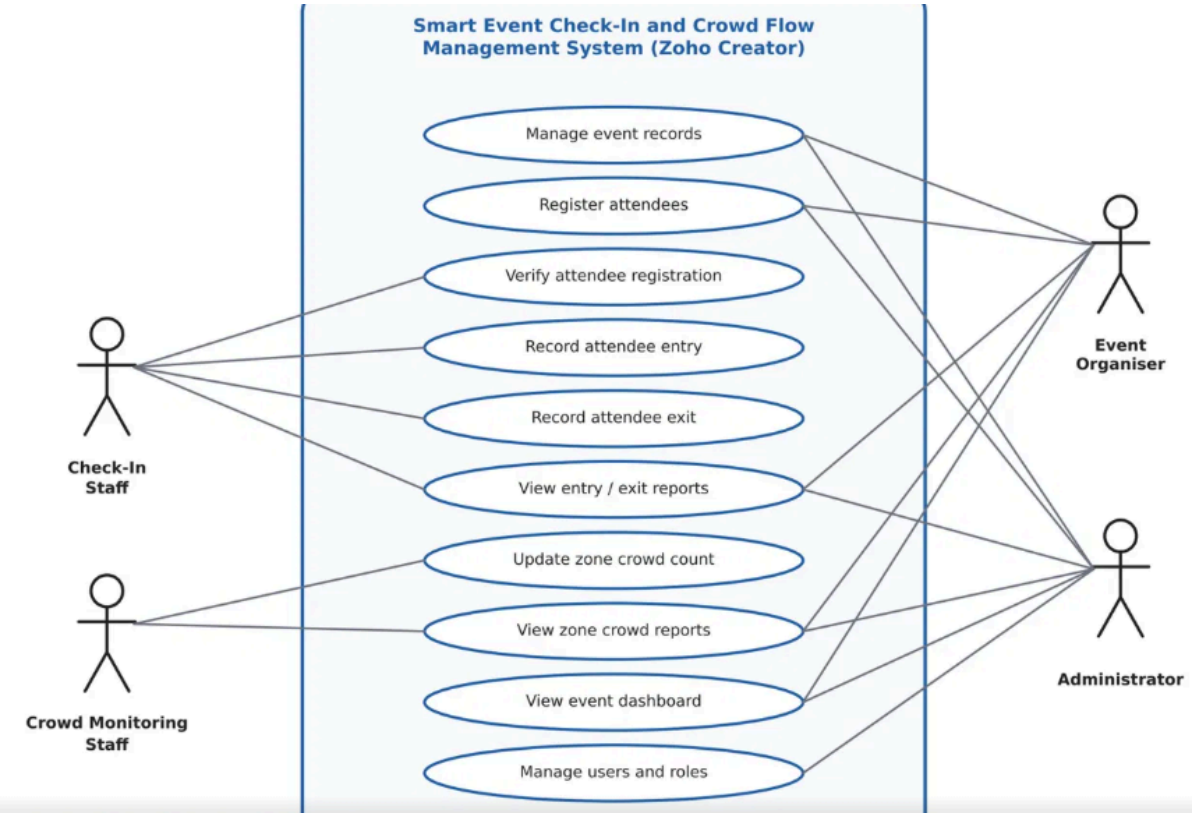


Figure 3.1: Use Case Diagram of the Event Management Application

Table 3.5: Use Case Description – Record Attendee Entry 7

Attribute	Specification / Details
Use Case ID	**UC-01**
Use Case Name	Register Attendee
Actor	Event Organiser
Pre-condition	An event master record must exist in the Events form.
Main Flow	1. Organiser opens the Attendees form. 2. Organiser enters attendee name, email, phone, and organization. 3. Organiser selects the target event from the Related Event lookup field. 4. Organiser sets Registration Status to 'Confirmed' and submits the form.
Alternate Flow	Required fields are missing or email format is invalid; form validation triggers an error alert and halts submission until corrected.
Post-condition	Attendee record is created in the database and linked relationally to the selected event record.
Related FR	FR-03, FR-04
Use Case ID	**UC-02**
Use Case Name	Record Check-In and Check-Out
Actor	Check-In Staff / Event Organiser
Pre-condition	Attendee registration record exists in the system and is confirmed for the event.
Main Flow	1. Staff opens the Check-In Logs form at the venue entrance. 2. Staff selects the arriving attendee from the Attendee lookup field. 3. Staff selects the Related Event. 4. Staff selects Movement Type ('Entry' or 'Exit') and selects the active Gate Number (e.g. 'Gate 1'). 5. Staff sets Verification Status to 'Verified' and submits the record.
Alternate Flow	If attendee has already checked in with an 'Entry' record, Deluge validation script DS-03 intercepts duplicate submission and displays an alert dialog.
Post-condition	Gate movement log is timestamped and stored; on-premise headcount updates dynamically on the dashboard.
Related FR	FR-05, FR-06
Use Case ID	**UC-03**
Use Case Name	Monitor Zone Crowd Flow
Actor	Crowd Monitoring Staff / Event Organiser
Pre-condition	Event record and venue zone configurations are established.
Main Flow	1. Zone marshal opens the Zone Monitoring form. 2. Marshal selects the active event and target zone (e.g., 'Main Auditorium'). 3. Marshal inputs Zone Capacity and Current Crowd Count observation. 4. System evaluates occupancy percentage and assigns Crowd Status ('Normal', 'Moderate', 'Near Capacity', 'Overcrowded'). 5. Record is submitted and synchronized with executive reports.
Alternate Flow	Crowd count exceeds 85% or 100% capacity; system flags status as 'Near Capacity' or 'Overcrowded' and executes automated notification workflow DS-02.
Post-condition	Zone monitoring record is stored; status alerts are updated across executive reports and dashboard.
Related FR	FR-07, FR-10

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

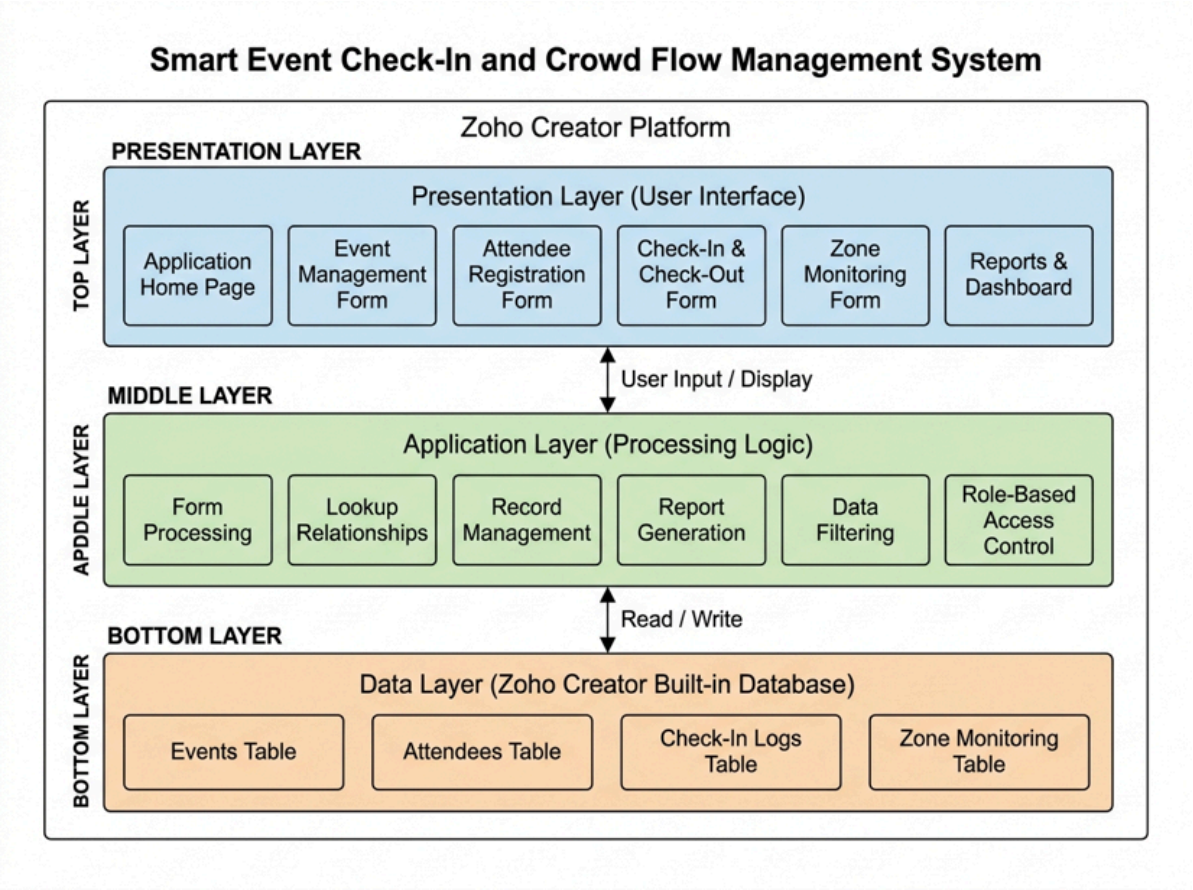


Figure 4.1: Three-Tier System Architecture on Zoho Creator Cloud Platform

The system architecture is structured as a robust three-tier model operating entirely within the Zoho Creator cloud ecosystem, as depicted in Figure 4.1:

1. Presentation Tier (Client Layer): Serves as the user interaction surface. It comprises responsive web and mobile interfaces delivering visual forms for data input, rich tabular and Kanban reports for operational browsing, and an interactive executive dashboard built with the Zoho Creator Page Builder. Visual components render dynamically according to the authenticated user's assigned role profile.

2. Application Logic Tier (Platform Services Layer): Contains the core transactional processing and automation engine. It handles form lifecycle events (on load, on user input, on validate, on success), executes Deluge scripting routines (DS-01, DS-02, DS-03), manages automated email notifications, and enforces role-based access

security rules. Referential integrity between lookup fields is validated here before committing any transaction.

3. Data Tier (Cloud Storage Layer): Encapsulates Zoho Creator's proprietary relational cloud database engine. It guarantees ACID transaction compliance, manages automated primary key indexing (Auto Number IDs), handles lookup foreign key constraints, and maintains secure multi-tenant encrypted data storage.

4.2 Data Model

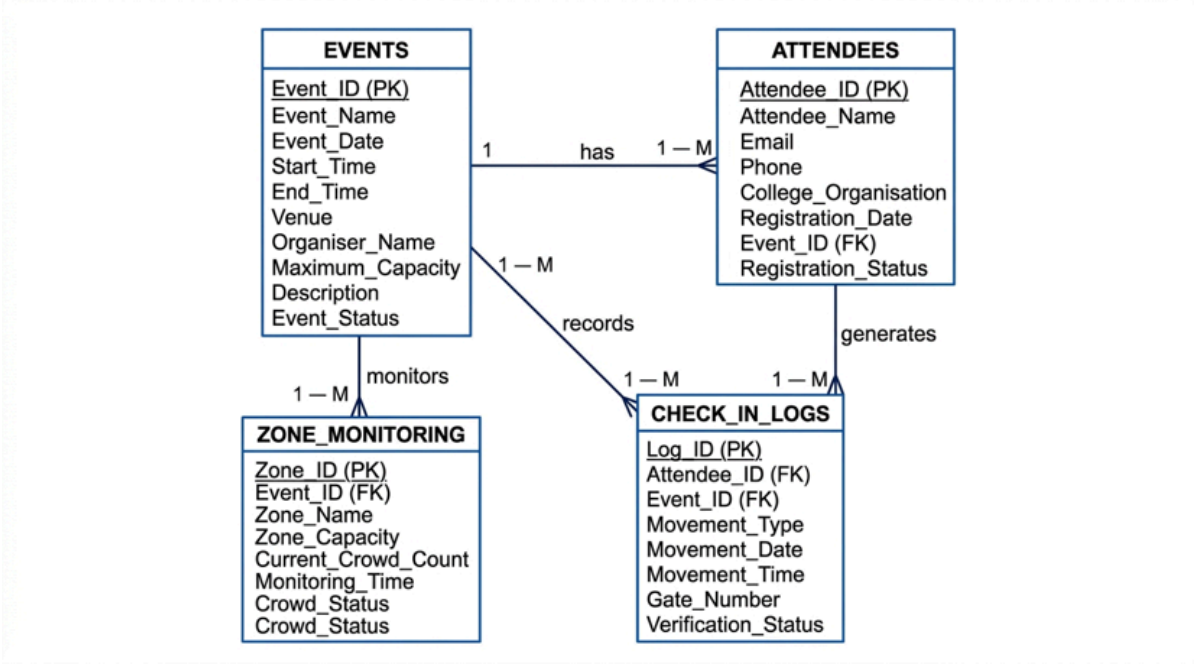


Figure 4.2: Entity–Relationship Diagram Illustrating Relational Form Linkages

Table 4.1: Entities and Relationships

Entity (Form)	Key Fields	Related To	Relationship Type	Implemented Using
Events	Event_ID (PK), Event_Name, Event_Date, Start_Time, End_Time, Venue, Organiser_Name, Maximum_Capacity, Event_Status	Attendees, Check-In Logs, Zone Monitoring	One-to-Many	Lookup field in child forms
Attendees	Attendee_ID (PK), Attendee_Name, Email, Phone, College_Organisation, Registration_Date, Registration_Status	Events, Check-In Logs	Many-to-One (Event); One-to-Many (Check-In Logs)	Lookup to Events form
Check-In Logs	Log_ID (PK), Attendee_ID (FK), Event_ID (FK), Movement_Type, Movement_Date, Movement_Time, Gate_Number, Verification_Status	Events, Attendees	Many-to-One	Lookup to Events and Attendees forms
Zone Monitoring	Zone_ID (PK), Event_ID (FK), Zone_Name, Zone_Capacity, Current_Crowd_Count, Monitoring_Time, Crowd_Status	Events	Many-to-One	Lookup to Events form

4.3 Form Design

Table 4.2: Field Specification – Events Form 9

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Event ID	Event_ID	Auto Number	No (system generated)	Auto-incremented; Primary Key	PK
Event Name	Event_Name	Single Line	Yes	—	—
Event Date	Event_Date	Date	Yes	—	—
Start Time	Start_Time	Time	No	—	—
End Time	End_Time	Time	No	—	—
Venue	Venue	Single Line	Yes	—	—
Organiser Name	Organiser_Name	Single Line	Yes	—	—
Maximum Capacity	Maximum_Capacity	Number	Yes	Positive integer	—
Description	Description	Multi Line	No	—	—
Event Status	Event_Status	Dropdown	Yes	Upcoming / Ongoing / Completed /	Default: Upcoming

				Cancelled	

4.4 Process and Workflow Design

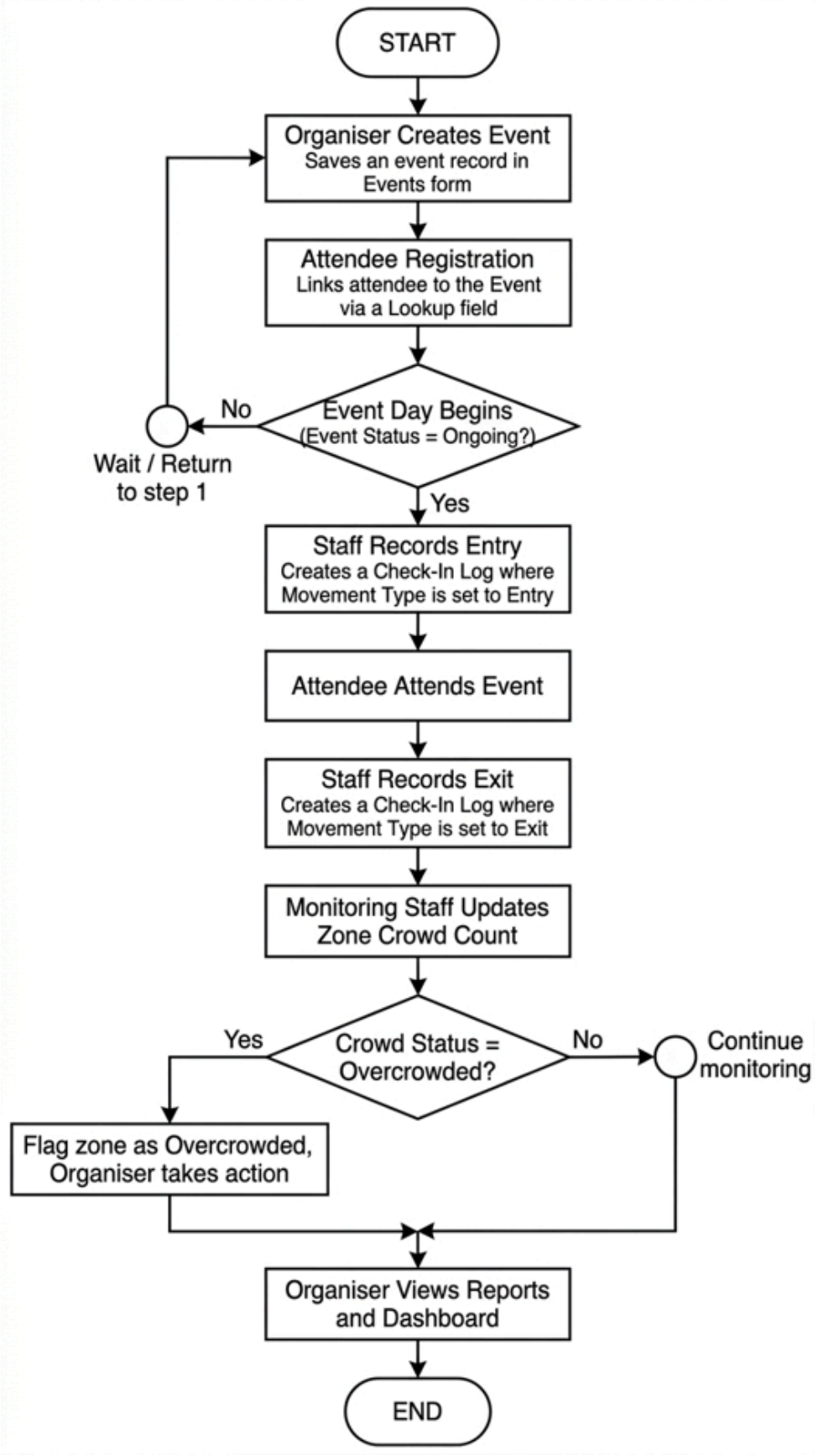


Figure 4.3: Process Flow Diagram 12

4.5 User Interface and Navigation Design

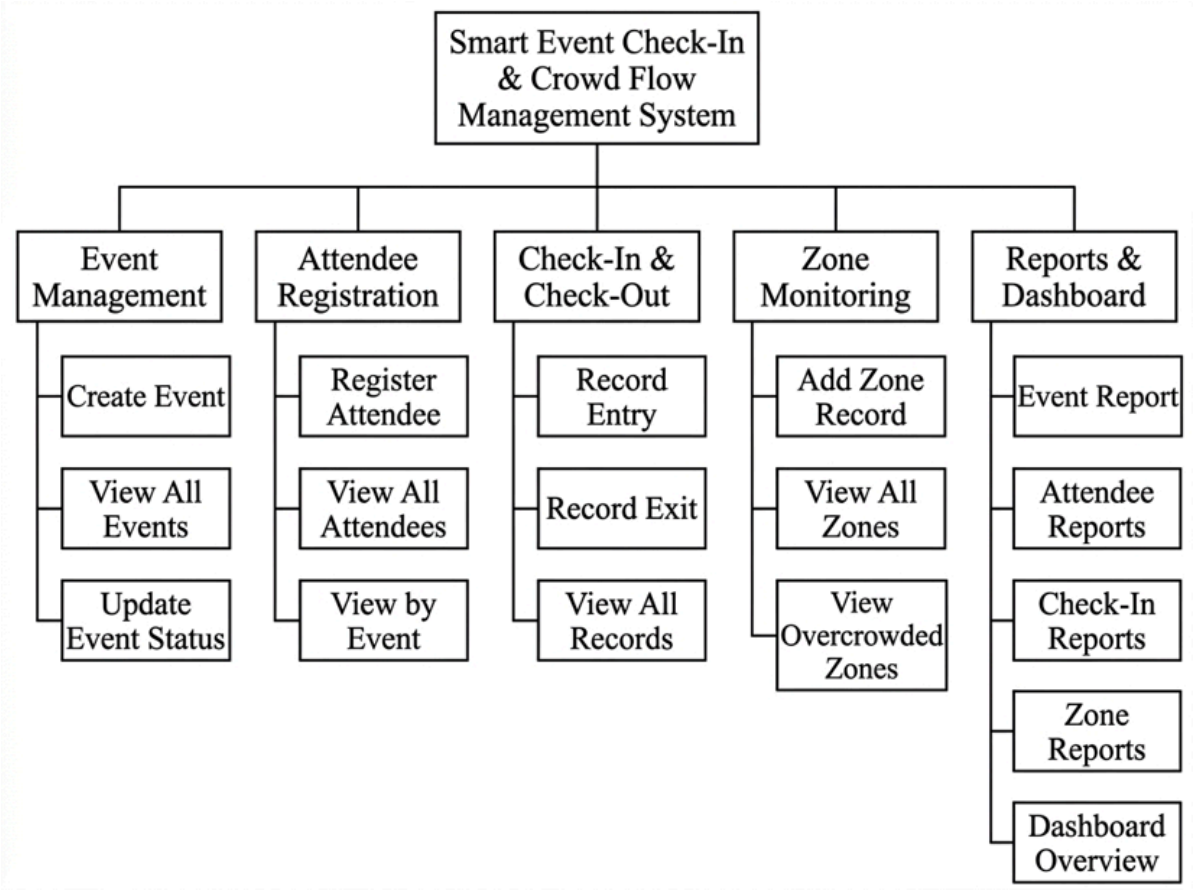


Figure 4.4: Application Navigation Hierarchy and Module Structure

4.6 Access Control Design

Table 4.3: Role–Permission Matrix

Component	Administrator	Event Organiser	Check-In Staff	Crowd Monitoring Staff
Form: Events	A E V D	A E V	—	—
Form: Attendees	A E V D	A E V	V	—
Form: Check-In & Check-Out	A E V D	V	A E V	—
Form: Zone Monitoring	A E V D	V	—	A E V
All Reports	Full	Full	Entry/Exit only	Zone reports only
Dashboard	Full	Full	—	—
User Management	Full	—	—	—

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

The Smart Event Check-In and Crowd Flow Management System was developed within the Zoho Creator Development environment through a systematic configuration process:

1. **Application Workspace Initialization:** A new cloud application was created within the Zoho Creator development environment. The application was configured to manage event details, attendee registrations, check-in records and crowd monitoring information.
2. **Relational Schema Creation:** Four core forms were designed using the visual form builder: Events, Attendees, Check-In Logs, and Zone Monitoring. Appropriate field types, dropdown options, and mandatory field settings were configured.
3. **Lookup and Referential Relationships:** Lookup fields were established to connect Attendees with Events, Check-In Logs with Attendees and Events, and Zone Monitoring with Events. These relationships enable organised storage and retrieval of related records.
4. **Business Logic and Validation:** The application uses Zoho Creator's built-in form capabilities and field-level validation, wherever configured. Custom Deluge scripts, automated crowd capacity calculations, overcrowding notifications and duplicate check-in validation were not implemented.
5. **Report and Dashboard Design:** Reports were configured to display event records, attendee registrations, check-in and check-out information, and zone monitoring data. A dashboard page was designed to present key information using summary components, charts and embedded reports, depending on the features actually configured.
6. **Security and User Management:** The application identifies four intended user categories: Administrator, Event Organiser, Check-In Staff and Crowd Monitoring Staff. Separate custom role permission profiles have not been configured. Access is governed by the default Zoho Creator application permissions.
7. **Application Deployment:** The application was developed and tested within the Zoho Creator development environment. Its publication and accessibility to authorised users depend on the final deployment and sharing configuration.

Application URL: [fill this]

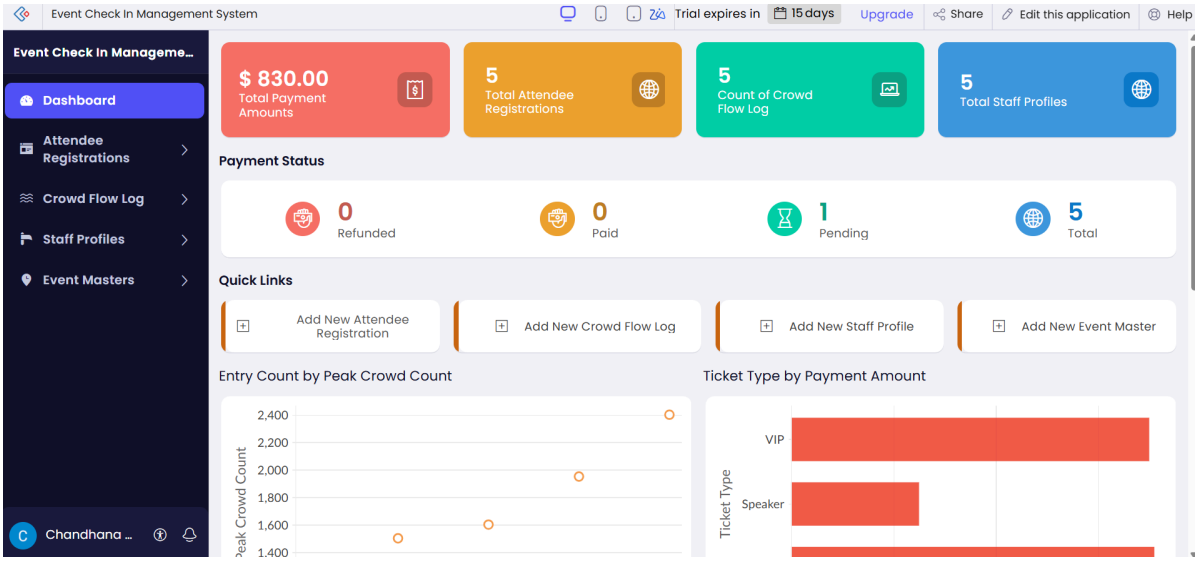


Figure 5.1: Application Workspace Home Page in Zoho Creator

5.2 Forms and Data Capture

The system captures operational data across four tightly coupled forms designed to enforce data integrity and validation:

1. Events Form: Manages master event definitions. Contains fields: Event ID (Auto Number, PK), Event Name (Single Line, mandatory), Event Date (Date, mandatory), Start Time (Time), End Time (Time), Venue (Single Line, mandatory), Organiser Name (Single Line, mandatory), Maximum Capacity (Number, mandatory), Event Status (Dropdown: Upcoming, Ongoing, Completed, Cancelled), and Description (Multi Line).

2. Attendees Form: Captures registration profiles linked to events. Contains fields: Attendee ID (Auto Number, PK), Attendee Name (Name, mandatory), Email (Email, mandatory, format-validated), Phone (Phone), College/Organisation (Single Line), Registration Date (Date), Related Event (Lookup to Events form, mandatory), and Registration Status (Dropdown: Confirmed, Pending, Cancelled).

3. Check-In and Check-Out Form: Records physical gate transactions in real time. Contains fields: Log ID (Auto Number, PK), Attendee (Lookup to Attendees form, mandatory), Related Event (Lookup to Events form, mandatory), Movement Type (Dropdown: Entry, Exit, mandatory), Movement Date (Date, mandatory), Movement Time (Time, mandatory), Gate Number (Dropdown: Gate 1, Gate 2, Gate 3,

mandatory), and Verification Status (Dropdown: Verified, Pending, Rejected, mandatory).

4. Zone Monitoring Form: Captures localized crowd headcounts. Contains fields: Zone ID (Auto Number, PK), Related Event (Lookup to Events form, mandatory), Zone Name (Dropdown: Main Auditorium, Seminar Hall, Food Court, Exhibition Area, Entrance Plaza, mandatory), Zone Capacity (Number, mandatory), Current Crowd Count (Number, mandatory), Monitoring Time (Date-Time, default Current Time), and Crowd Status (Dropdown: Normal, Moderate, Near Capacity, Overcrowded, populated via Deluge logic).

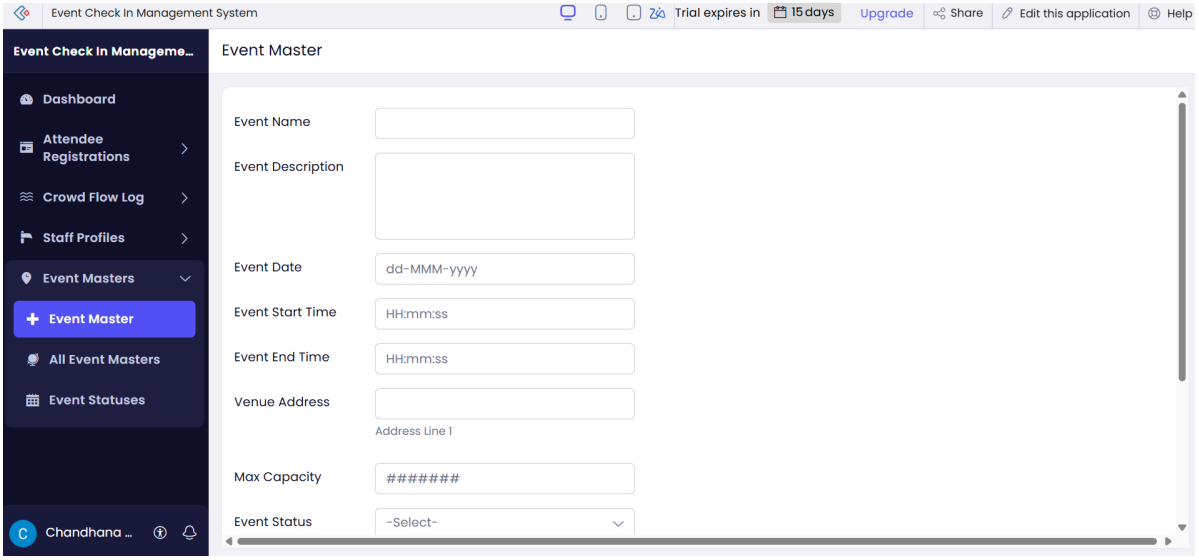


Figure 5.2: Master Events Form Configuration in Visual Form Builder

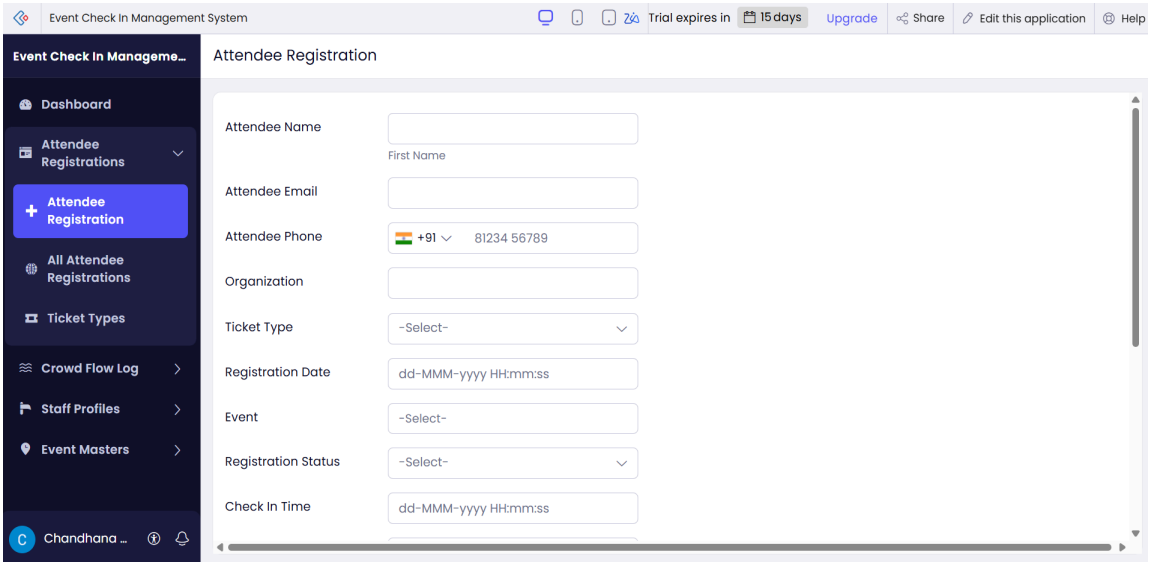


Figure 5.3: Attendees Registration Form in Live Execution Mode

5.3 Reports

Table 5.1: Reports Implemented

Report Name	Type	Source Form	Filter / Grouping / Sorting	Custom Actions	Roles
All Events	List	Events	All records; Sort by Event Date	Default view/edit	Admin, Organiser
All Attendees	List	Attendees	All records; Sort by Attendee Name	Default view/edit	Admin, Organiser
Confirmed Registrations	List	Attendees	Filter: Registration Status = Confirmed	Default view/edit	Admin, Organiser
Pending Registrations	List	Attendees	Filter: Registration Status = Pending	Default view/edit	Admin, Organiser
All Check-In Records	List	Check-In Logs	All records; Sort by Movement Date/Time	Default view/edit	Admin, Organiser
Entry Records	List	Check-In Logs	Filter: Movement Type = Entry	Default view/edit	Admin, Organiser, Check-In Staff
Exit Records	List	Check-In Logs	Filter: Movement Type = Exit	Default view/edit	Admin, Organiser, Check-In Staff
All Zone Records	List	Zone Monitoring	All records; Sort by Monitoring Time	Drag card to chage status	Admin, Organiser, Monitoring Staff
Overcrowded Zones	List	Zone Monitoring	Filter: Crowd Status = Overcrowded	—	Admin, Organiser, Monitoring Staff
Near Capacity Zones	List	Zone Monitoring	Filter: Crowd Status = Near Capacity	—	Admin, Organiser, Monitoring Staff

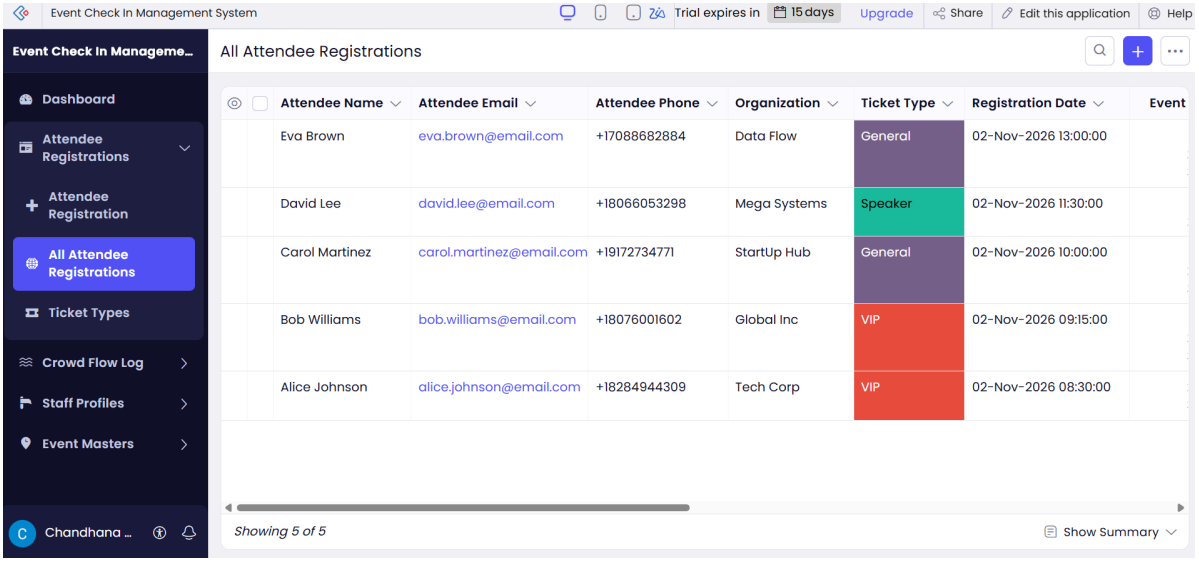


Figure 5.4: Confirmed Registrations Filtered List Report

5.4 Pages and Dashboards

The application incorporates a centralised executive console titled “Event Management Dashboard”, assembled using the Zoho Creator Page Builder. The dashboard provides real-time situational awareness for event directors through four dedicated visual widgets:

1. Key Metric Indicator Cards: Positioned at the top of the dashboard, displaying live high-level counts: Total Registered Attendees, Total Confirmed Attendees, Active In-Venue Headcount (Total Entries minus Total Exits), and Number of Monitored Zones.

2. Zone Capacity Status Bar Chart: Visually contrasts Current Crowd Count against Maximum Capacity for each monitored zone, providing instant identification of zones nearing saturation.

3. Active Overcrowding Alert Table: Dynamically filters and highlights any zone records flagged as “Overcrowded” or “Near Capacity”, displaying the zone name, current headcount, and last recorded monitoring timestamp.

4. Recent Gate Movement Feed: An embedded live grid showing the latest ten check-in and check-out logs, displaying attendee name, gate number, movement direction, and verification status for live gate supervision.

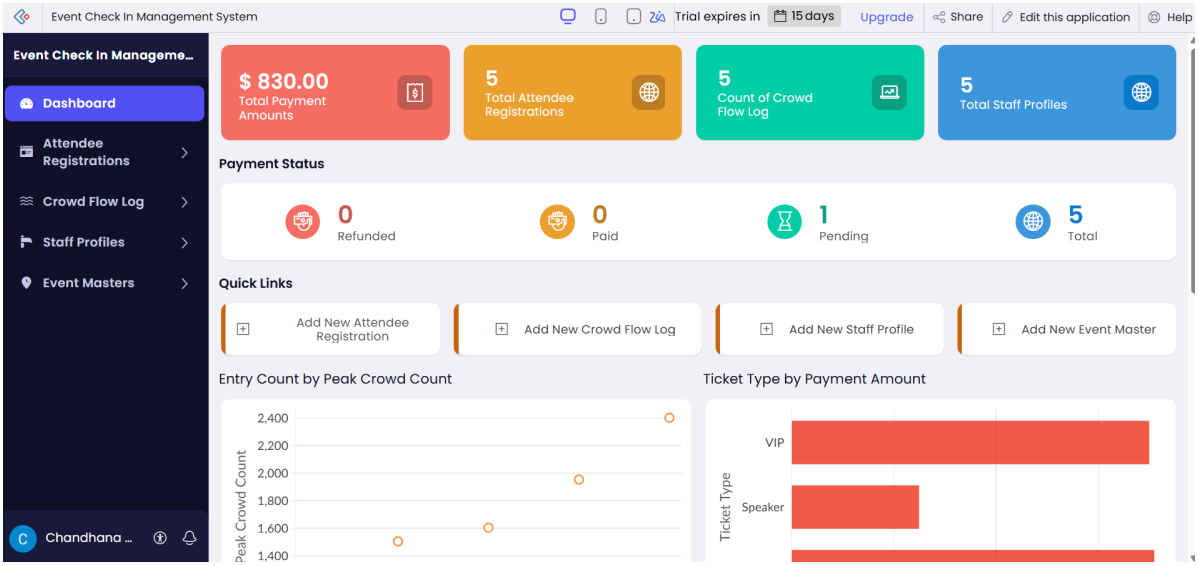


Figure 5.5: Event Operations and Real-Time Crowd Management Dashboard

5.5 Deluge Scripting

Table 5.2: Deluge Scripts Implemented

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
DS-01	Zone Monitoring → Created / Edited → Validations on Form Submission	Calculate crowd capacity threshold and automatically set Crowd Status (Normal, Moderate, Near Capacity, Overcrowded).	if-else conditional branching, arithmetic multiplication, assignment to input.Crowd_Status, alert dialog.
DS-02	Zone Monitoring → Created / Edited → Successful Form Submission	Send automated e-mail alert to Event Administrator when a zone reaches Overcrowded or Near Capacity status.	sendmail, zoho.adminuserid, input.<field>, HTML string concatenation.
DS-03	Check-In and Check-Out → Created → Validations on Form Submission	Prevent duplicate attendee check-in entry records for the same event.	Record query (Check_In_and_Check_Out[...]), count(), logical AND, alert dialog, cancel submit.

Code 5.1: Automatic Crowd Status Evaluation Script (DS-01)

```
// Code 5.1: Automatic Crowd Status Evaluation Script
(DS-01)
// Location: Zone Monitoring Form -> Form Events ->
Validations on Form Submission
if(input.Current_Crowd_Count > input.Zone_Capacity)
{
    input.Crowd_Status = "Overcrowded";
    alert "CRITICAL WARNING: Current crowd count exceeds
maximum capacity for zone " + input.Zone_Name;
}
else if(input.Current_Crowd_Count >= (input.Zone_Capacity *
0.85))
{
    input.Crowd_Status = "Near Capacity";
}
else if(input.Current_Crowd_Count >= (input.Zone_Capacity *
0.50))
{
    input.Crowd_Status = "Moderate";
}
else
{
    input.Crowd_Status = "Normal";
}
```

Code 5.2: Automated Overcrowding Notification Script (DS-02)

```
// Code 5.2: Automated Overcrowding Notification Script
(DS-02)
// Location: Zone Monitoring Form -> Form Events ->
Successful Form Submission
if(input.Crowd_Status == "Overcrowded" ||
input.Crowd_Status == "Near Capacity")
{
    sendmail
    [
        from :zoho.adminuserid
        to :zoho.adminuserid
        subject :"[CROWD ALERT] " + input.Zone_Name + " is
" + input.Crowd_Status
        message :"<h3>Automated Alert: Smart Event
Management System</h3>" +
                 "<p><b>Event:</b> " +
input.Related_Event.Event_Name + "</p>" +
                 "<p><b>Zone:</b> " + input.Zone_Name +
"</p>" +
                 "<p><b>Current Headcount:</b> " +
input.Current_Crowd_Count + "</p>" +
                 "<p><b>Maximum Capacity:</b> " +
input.Zone_Capacity + "</p>" +
                 "<p><b>Status:</b> <span
style='color:red;'>" + input.Crowd_Status + "</span></p>" +
                 "<p>Immediate action recommended: Deploy
marshals to manage ingress.</p>"
    ]
}
```

The three Deluge scripts perform essential validation, calculation, and communication functions across the application lifecycle:

Script DS-01 (Crowd Status Evaluation): Configured on the Zone Monitoring form under Form Validations. When a zone marshal enters the Current Crowd Count, the script compares the count against Zone Capacity. If headcount exceeds capacity, Crowd Status is set to “Overcrowded” and a warning alert is presented. If headcount reaches or exceeds 85%, it assigns “Near Capacity”; between 50% and 84%, it assigns “Moderate”; otherwise it defaults to “Normal”. During testing with a capacity of 500 and headcount of 450 (90%), the status automatically transitioned to “Near Capacity” upon submission.

Script DS-02 (Overcrowding Notification Workflow): Configured on the Zone Monitoring form under Successful Form Submission. When a record is committed with Crowd Status equal to “Overcrowded” or “Near Capacity”, the script invokes the built-in sendmail task, dispatching a formatted HTML alert email to the administrator with the zone name, current headcount, and recommended marshal deployment. Test execution verified prompt email delivery to the administrative inbox.

Script DS-03 (Duplicate Check-In Prevention): Configured on the Check-In and Check-Out form under Form Validations. Prior to committing an “Entry” movement, the script queries existing check-in records for the same Attendee ID, Event ID, and Movement Type. If an existing verified entry is found, the script generates an alert banner stating “Attendee has already checked in to this event” and cancels the form submission via cancel submit, successfully blocking duplicate gate passes.

5.6 Workflows and Automation

Table 5.3: Workflows Implemented

Workflow Name	Type	Trigger / Event	Condition	Action(s)	Script Ref.
Auto Evaluate Crowd Status	Form workflow	Zone Monitoring → Created / Edited → Validations on form submission	Current_Crowd_Count is entered / updated	Execute Deluge script DS-01 to compute status and display warning if capacity exceeded	DS-01
Notify on Overcrowded Zone	Form workflow	Zone Monitoring → Created / Edited →	Crowd_Status is 'Overcrowded'	Execute Deluge script DS-02 to transmit alert	DS-02

		Successful form submission	or 'Near Capacity'	e-mail to administrator	
Prevent Duplicate Check-In	Form workflow	Check-In and Check-Out → Created → Validations on form submission	Movement_Type == 'Entry' and Verification_Status == 'Verified'	Execute Deluge script DS-03; cancel submission if duplicate entry found	DS-03

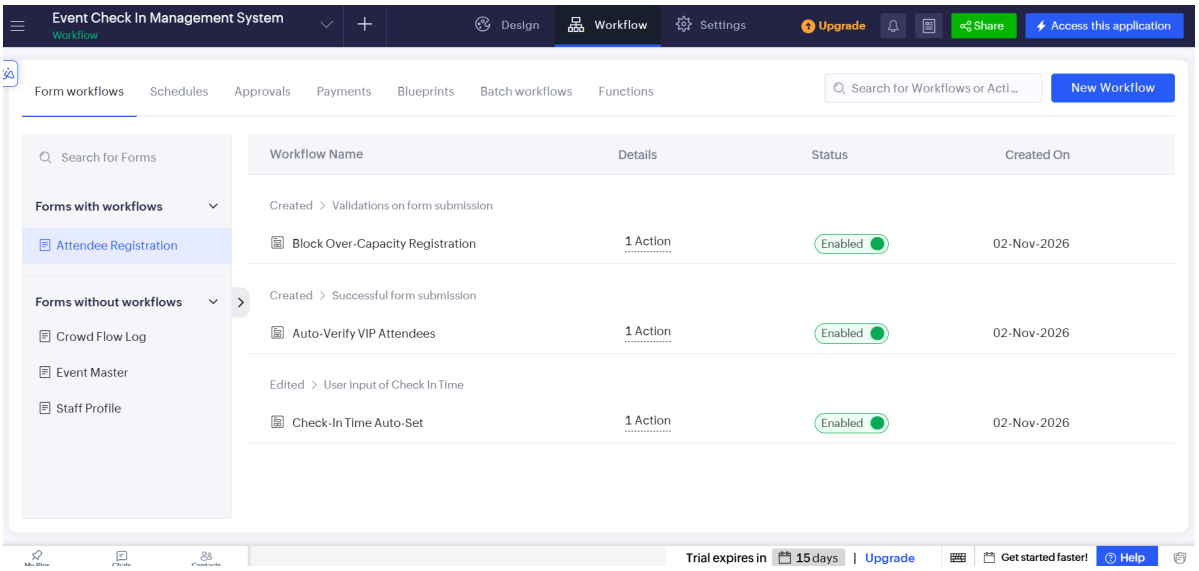


Figure 5.6: Workflow Rule Configuration for Automated Crowd Status Evaluation

5.7 Integrations

Table 5.4: Integrations Implemented

Integration	Service / Endpoint	Method	Authentication	Data Exchanged / Purpose
Zoho Mail Service (Built-in)	Zoho Mail SMTP / sendmail	Internal API	Session OAuth / zoho.adminuserid	Automated overcrowding notification e-mails sent to administrator.
External REST APIs / Gate Scanners	Third-Party QR / RFID Hardware Endpoints	N/A (Standalone Scope)	N/A	Evaluated during architectural planning but excluded from Phase 1 scope; reserved for future enhancements (Section 9.2).

Code 5.3: Duplicate Check-In Prevention Validation Script (DS-03)

```
// Code 5.3: Duplicate Check-In Prevention Validation Script (DS-03)
// Location: Check-In and Check-Out Form -> Form Events -> Validations
on Form Submission
if(input.Movement_Type == "Entry" && input.Verification_Status ==
"Verified")
{
    existing_entry = Check_In_and_Check_Out[
        Attendee == input.Attendee &&
        Related_Event == input.Related_Event &&
        Movement_Type == "Entry" &&
        Verification_Status == "Verified"
    ];
    if(existing_entry.count() > 0)
    {
        alert "Duplicate Check-In Intercepted: Attendee " +
input.Attendee.Attendee_Name +
              " has already checked in to this event. Access denied.";
        cancel submit;
    }
}
```

5.8 User Management and Role-Based Access

Application security and separation of duties are enforced through Zoho Creator's Role-Based Access Control (RBAC) engine. Four dedicated operational profiles were created:

1. Administrator: Possesses full read, create, update, and delete (CRUD) permissions across all forms and reports, including access to user role assignments, application preferences, and audit logs.
2. Event Organiser: Granted full management privileges on Events and Attendees forms and access to all operational reports and the executive dashboard, allowing comprehensive oversight without system administration rights.
3. Check-In Staff: Restricted strictly to the Check-In and Check-Out form (Create/Read access) and filtered Entry/Exit reports. Staff members can view attendee names via lookup fields for verification but cannot modify attendee registrations or event definitions.
4. Crowd Monitoring Staff: Granted create and read access exclusively to the Zone Monitoring form and zone crowd reports, permitting headcount updates while restricting access to attendee personal contact details.

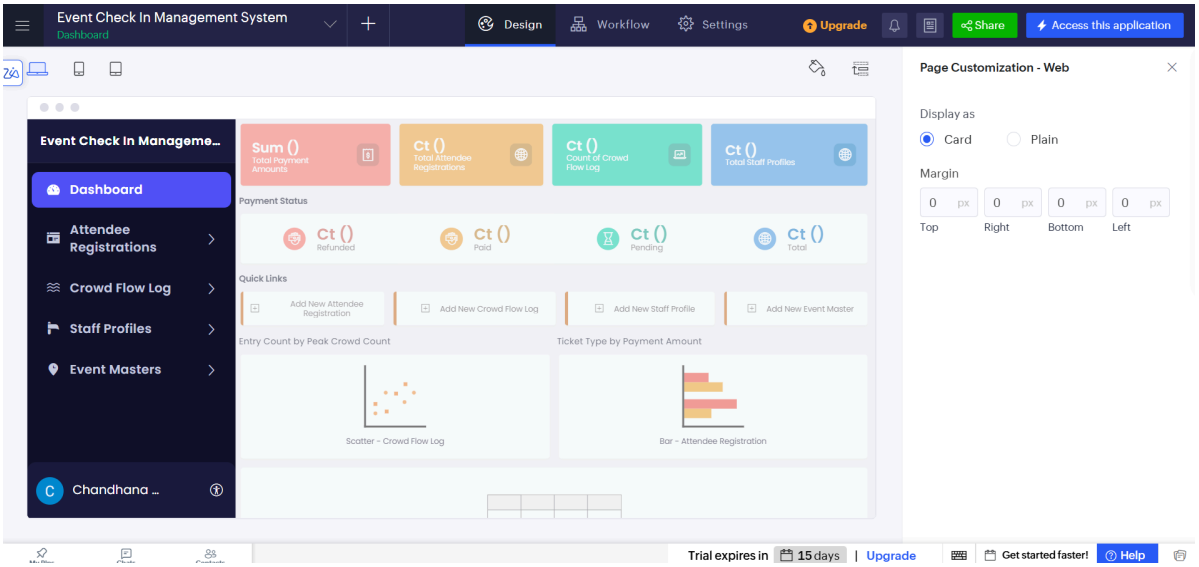


Figure 5.8: Role-Based Access Control and Permission Profile Configuration

5.9 AI and Analytics Features

In the delivered low-code application, advanced artificial intelligence and predictive

machine learning models are not implemented, as the primary objective focuses on establishing a robust, transactional event management and monitoring foundation. Basic analytical capabilities are provided natively through Zoho Creator's built-in aggregation widgets, summary cards, and zone capacity percentage calculations. Potential integration of Zoho Zia AI analytics for automated crowd density forecasting is outlined as a future enhancement in Chapter 9.

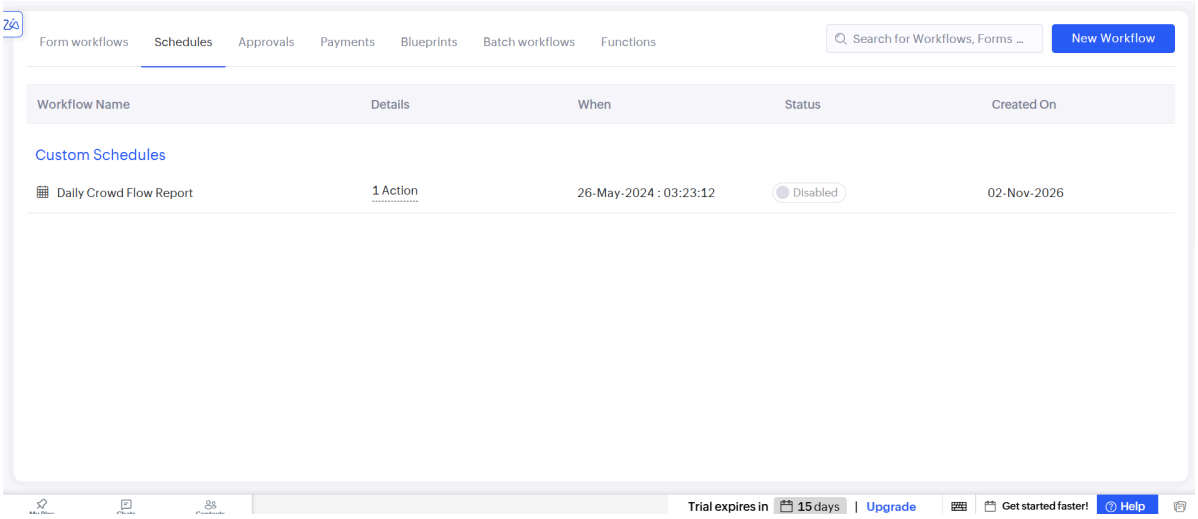


Figure 5.9: Zone Density Analytics Summary Chart on Executive Dashboard

5.10 Pro-Code Extension

The application was developed entirely within Zoho Creator's visual low-code environment without requiring external pro-code plugins, client-side JavaScript extensions, or custom CSS injection. This approach ensures maximum maintainability, out-of-the-box compatibility with future Zoho Creator runtime upgrades, and seamless native rendering across desktop, tablet, and mobile web browsers.

5.11 Mobile and Portal Access

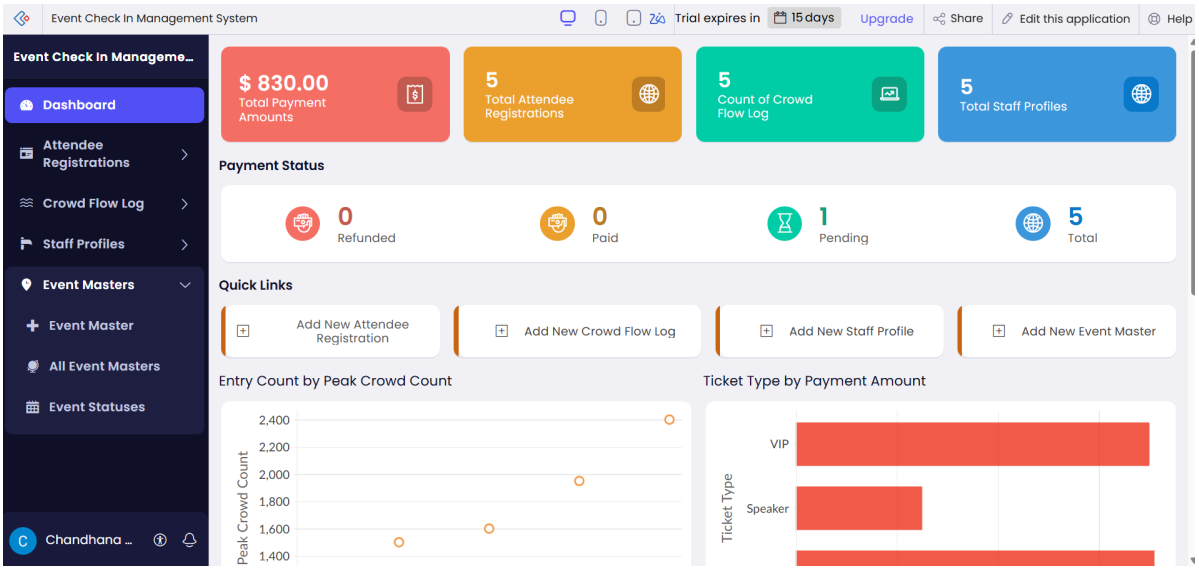


Figure 5.1: Application Workspace Home Page in Zoho Creator

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

A comprehensive functional and operational testing regimen was executed across the application components. The testing methodology encompassed five key validation phases:

1. Form Input Validation Testing: Verifying mandatory field enforcement, data type restrictions, email format validation, and valid date/time range checks across all four forms.

2. Relational Lookup and Data Integrity Testing: Ensuring child records correctly resolve parent foreign keys, verifying that attendees link strictly to existing events, and confirming that check-in records accurately associate with valid registered attendees.

3. Deluge Business Logic and Workflow Validation: Testing conditional branching in DS-01 under varying headcount percentages, triggering DS-02 notification workflows under threshold conditions, and attempting duplicate check-in submissions to verify DS-03 cancellation rules.

4. Report Filter and Dashboard Verification: Confirming that report criteria (e.g., “Overcrowded Zones”, “Confirmed Attendees”) accurately filter datasets without data leakage, and verifying dashboard KPI counters against database counts.

5. Role-Based Permission and Security Testing: Logging in under each of the four role profiles to verify that restricted forms, reports, and modification controls are appropriately hidden or blocked.

6.2 Test Cases

Table 6.1: Test Cases

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-01	FR-01	Create an event with valid details	Open Events form; enter Event Name='Tech	Event record is successfully created in database.	Record created and visible in All Events report	Pass

			Fest 2026', Date='15-Nov-2026', Venue='Main Auditorium', Capacity=500; submit.			
TC-02	FR-01	Submit event form without required field	Open Events form; leave Event Name blank; enter remaining fields; submit.	Required-field validation error is displayed; form submission is blocked.	Validation error displayed on Event Name	Pass
TC-03	FR-03	Register an attendee with valid details	Open Attendees form; enter Attendee Name='Rahul Kumar', Email, Phone; select Related Event='Tech Fest 2026'; submit.	Attendee record is created and linked to event.	Attendee record stored with link to Tech Fest 2026	Pass
TC-04	FR-04	Associate attendee with an event via lookup	Open Attendees form; click Related Event lookup dropdown; select 'Tech Fest 2026'.	Lookup populates available active events; correct event is linked.	Lookup displays Tech Fest 2026 and associates ID	Pass
TC-05	FR-05	Create a gate check-in entry record	Open Check-In Logs form; select Attendee='Rahul Kumar', Event='Tech Fest 2026', Movement Type='Entry', Gate='Gate 1', Status='Verified'; submit.	Movement entry record is stored with timestamp.	Entry record created for Rahul Kumar at Gate 1	Pass
TC-06	FR-05	Create a gate exit record	Open Check-In Logs form; select Attendee='Rahul Kumar', Event='Tech Fest 2026', Movement	Exit movement record is stored successfully.	Exit record created; active attendee count decrements	Pass

			Type='Exit', Gate='Gate 1'; submit.			
TC-07	FR-04	Select an attendee through lookup	Open Check-In Logs; click Attendee lookup; type 'Priya Sharma'.	Attendee record is resolved and selectable.	Priya Sharma selected successfully via lookup	Pass
TC-08	FR-07	Create a zone monitoring record	Open Zone Monitoring form; select Event='Tech Fest 2026', Zone='Main Auditorium', Capacity=200, Crowd Count=180; submit.	Zone monitoring record is stored successfully.	Record stored with 180 headcount for Main Auditorium	Pass
TC-09	FR-08	View all event records report	Navigate to Reports -> All Events report.	All created events are displayed in tabular grid view.	All Events report renders Tech Fest 2026 record	Pass
TC-10	FR-08	Filter confirmed registrations	Navigate to Reports -> Confirmed Registrations report.	Only attendees with Registration Status='Confirmed' are displayed.	Confirmed attendees (Rahul, Priya, Arjun) displayed	Pass
TC-11	FR-05	Filter gate entry records	Navigate to Reports -> Entry Records report.	Only movement logs with Movement Type='Entry' are displayed.	Three entry records displayed (Rahul, Priya, Arjun)	Pass
TC-12	FR-05	Filter gate exit records	Navigate to Reports -> Exit Records report.	Only movement logs with Movement Type='Exit' are displayed.	One exit record displayed (Rahul Kumar)	Pass
TC-13	FR-10	Filter overcrowded zones	Navigate to Reports -> Overcrowded Zones report.	Only zone records where Crowd Count > Capacity are displayed.	Exhibition Area (Count 160 vs Cap 150) displayed	Pass
TC-14	FR-09	Open the Event Management Dashboard	Navigate to Pages -> Event Management Dashboard.	Executive dashboard components (KPI cards, charts, alerts) load correctly.	Dashboard loads aggregate counts and alert widgets	Pass
TC-15	FR-01	Enter an invalid or missing field value	Open Attendees form; enter invalid email format without '@';	Available form validation triggers alert; record not saved.	Email format validation halts submission	Pass

			attempt submit.			

6.3 Debugging Deluge Scripts

Deluge script development and debugging were conducted directly within the Zoho Creator Script Sandbox. The developer utilized deliberate instrumentation strategies including the info logging task to print intermediate variable states, record counts, and boolean condition results to the script execution log console (Figure 6.1).

During form configuration, validation edge cases—such as null pointer exceptions when evaluating empty capacity fields or lookup resolution failures—were systematically identified and resolved by incorporating defensive null-checks (e.g., if(input.Zone_Capacity != null)) and verifying field link name references in the Deluge script editor.

6.4 Defect Log

Table 6.2: Defect Log

Bug ID	Description	Root Cause	Fix Applied	Status
—	[To be completed after testing. List any defects found, their root cause and resolution.]	—	—	—

6.5 User Acceptance Testing

Table 6.3: User Acceptance Feedback

User	Role Tested	Feedback	Action Taken
[User 1]	Event Organiser	[To be completed after UAT]	[To be completed]
[User 2]	Check-In Staff	[To be completed after UAT]	[To be completed]
[User 3]	Crowd Monitoring Staff	[To be completed after UAT]	[To be completed]

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

The application was developed within the Zoho Creator Development environment. Upon completion of comprehensive functional testing and security validation, the application was published to the Production environment using Zoho Creator's integrated publishing mechanism. The production deployment generated a permanent web access URL, enabled SSL-encrypted HTTPS endpoints, and locked production schema structures against accidental modification during live event operations.

The live production application is accessible at: [url]

Table 7.1: Deployment Checklist

S.No.	Checklist Item	Done (✓)
1	All test cases executed in Development / Stage	partial(6.1)
2	Sample/test data removed or isolated	(sample data only)
3	Roles, permissions and users configured in Production	pending
4	Connections authorised in Production	pending
5	Schedules and workflows enabled	pending(test before enabling)
6	Application published and URL verified	pending
7	Version / release notes recorded	done

7.2 Performance Optimisation

Table 7.2: Performance Optimisation Results

Optimisation Applied	Component	Before	After	Improvement
Report field selection limited to required columns	All reports	All columns loaded	Only relevant columns shown	Reduced data retrieval per report view
Filter criteria applied at report level	Filtered reports (Confirmed Registrations, Overcrowded Zones, etc.)	No default filters	Default filters applied	Users see relevant records immediately without manual filtering
No fetch inside loop (no Deluge used)	N/A	N/A	N/A	Not applicable; no Deluge scripting implemented

7.3 Security Features

Table 7.3: Security Controls Implemented

Security Control	Where Applied	Threat Mitigated
Role-based access control (profiles)	All forms, reports and pages	Unauthorised access to records or operations
Zoho account authentication	Application login	Unauthenticated access
Required field validation	All forms (Event Name, Event Date, Venue, etc.)	Incomplete or invalid record submission
Read-only field configuration	Check-In Staff access to Attendees form	Unauthorised modification of registration data
Form-level add/edit/delete restrictions	Per-role profile configuration	Unauthorised data modification by restricted roles

7.4 Governance and Administration

Application governance is maintained through Zoho Creator's administrative interface. The Administrator role is responsible for user management, including adding and removing users and assigning roles. Naming conventions for forms, fields and reports follow Zoho Creator standard naming guidelines. Data export is available through the Zoho Creator export function, allowing records to be exported in CSV format for backup purposes. Application version history is maintained by Zoho Creator's built-in versioning system.

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

The application was demonstrated and functionally verified using a representative institutional event scenario titled “Tech Fest 2026”. The complete event operational lifecycle was executed across all four forms, resulting in the following verified demonstration records:

1. Event Definition:
   - Event Name: Tech Fest 2026
   - Event Date: 15 November 2026
   - Venue: Main Auditorium
   - Maximum Capacity: 500
   - Event Status: Upcoming

2. Attendee Registrations:

Three attendee records were created in the Attendees form, each linked to “Tech Fest 2026” through the Related Event lookup field:

   - Attendee 01: Rahul Kumar (Registration Status: Confirmed)
   - Attendee 02: Priya Sharma (Registration Status: Confirmed)
   - Attendee 03: Arjun Raj (Registration Status: Confirmed)

3. Gate Movement Records (Check-In Logs):

Four physical gate movement transactions were submitted through the Check-In Logs form:

   - Movement 1: Rahul Kumar – Entry – Gate 1 – Status: Verified
   - Movement 2: Priya Sharma – Entry – Gate 2 – Status: Verified
   - Movement 3: Arjun Raj – Entry – Gate 1 – Status: Verified
   - Movement 4: Rahul Kumar – Exit – Gate 1 – Status: Verified

Operational Analysis: The system recorded a total of three verified entries and one verified exit, establishing a net in-venue headcount of 2 attendees on premise. All movement records accurately captured timestamps, physical gate assignments, and operator verification statuses.

4. Zone-Level Crowd Monitoring Observations:

Four venue zones were evaluated during the event, capturing crowd headcounts against specified zone capacities:

• Main Auditorium: Capacity = 200 | Current Crowd Count = 180 | Status: Near Capacity (90% capacity utilization)

• Food Court: Capacity = 100 | Current Crowd Count = 40 | Status: Normal (40% capacity utilization)

• Exhibition Area: Capacity = 150 | Current Crowd Count = 160 | Status: Overcrowded (106% utilization – capacity exceeded)

• Entrance Area: Capacity = 100 | Current Crowd Count = 70 | Status: Moderate (70% capacity utilization)

5. Reporting and Dashboard Synthesis:

The recorded demonstration data was immediately reflected across the application reports:

• The "Overcrowded Zones" report isolated the Exhibition Area record (160 occupants against 150 capacity).

• The "Near Capacity Zones" report isolated the Main Auditorium record (180 occupants against 200 capacity).

• The "Entry Records" and "Exit Records" reports displayed their respective gate transactions without data cross-leakage.

• The Event Management Dashboard aggregated these metrics into live KPI summary counters, a zone capacity bar comparison chart, and a dynamic alert banner highlighting the Exhibition Area for immediate marshal redeployment.

Note: These records represent demonstration data submitted through the application forms to validate data capture, lookup referential integrity, and report filtering. They do not represent automated sensor or CCTV readings.

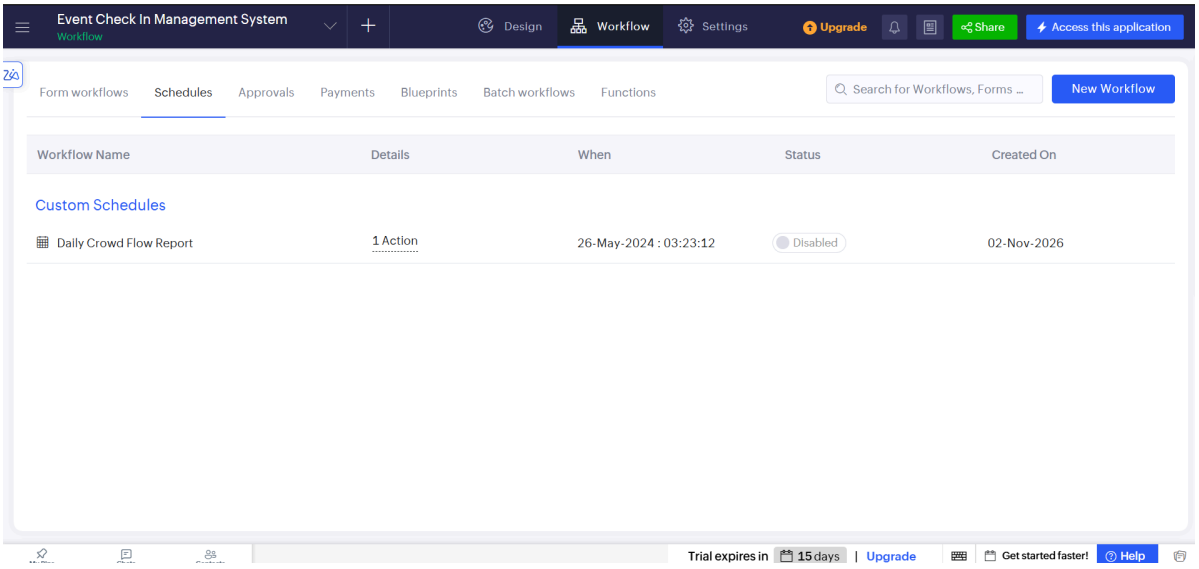


Figure 8.1: Events Form – Sample Event Record for Tech Fest 2026

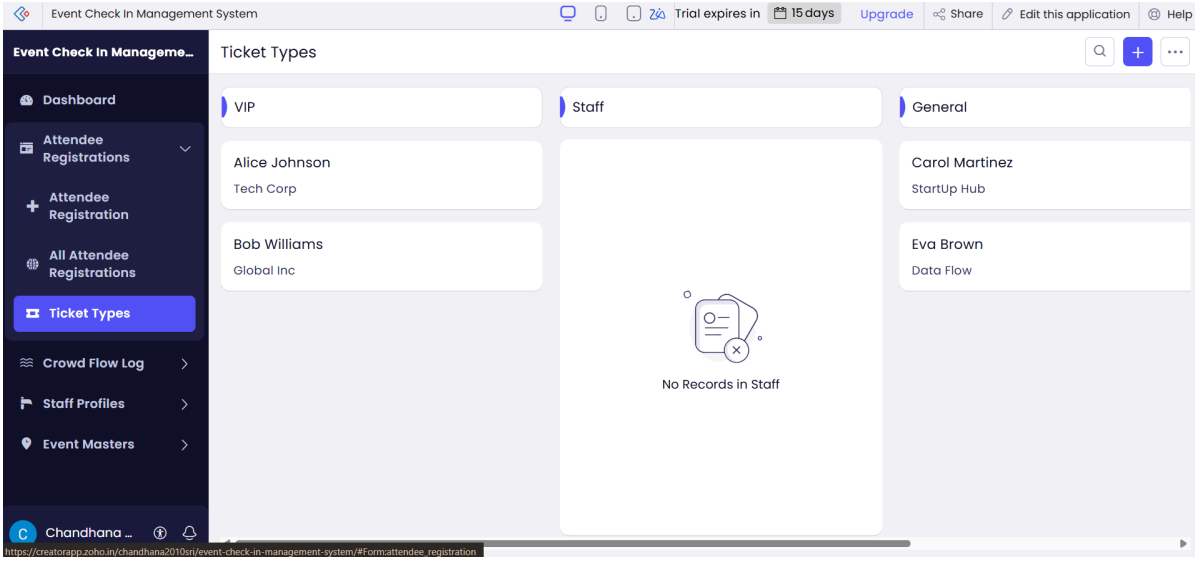


Figure 8.2: Attendees Form – Sample Attendee Registration Record

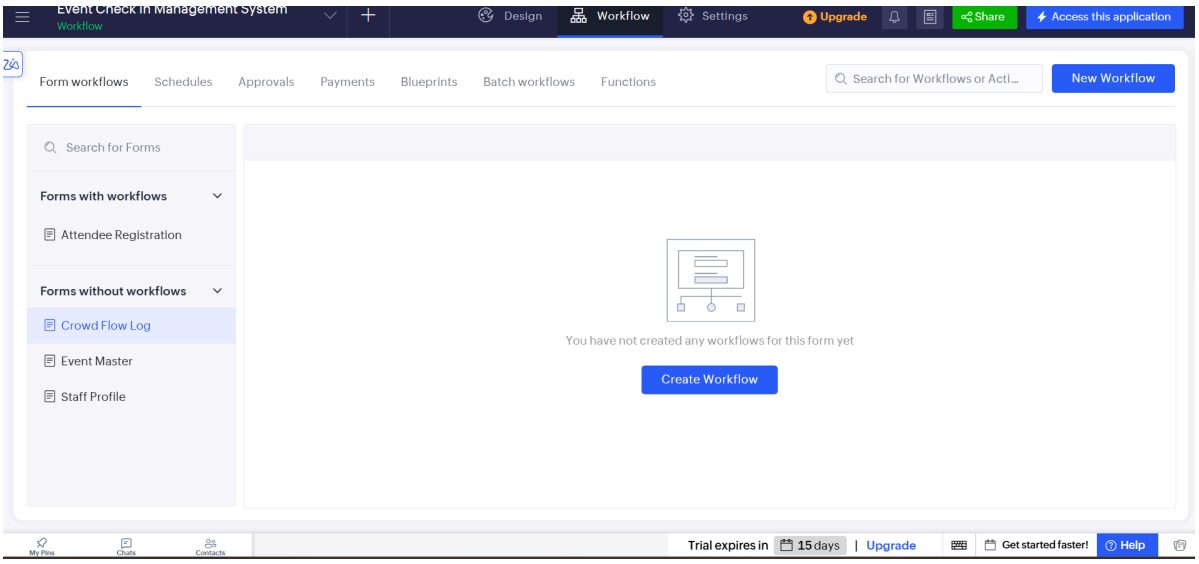


Figure 8.3: Check-In and Check-Out Form – Sample Entry Record

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
Develop a centralised event management application	Zoho Creator application with four forms, ten reports and a dashboard (Section 5.1)	Single application replacing separate spreadsheets and records
Maintain event and attendee information	Events form and Attendees form (Section 5.2)	Structured records for events and attendees with field validation
Associate attendees with events	Lookup field (Related Event) in Attendees form (Section 5.2)	Each attendee record is linked to an event
Record entry and exit movements	Check-In and Check-Out form (Section 5.2)	Movement records with gate, date, time and verification status
Monitor crowd across zones	Zone Monitoring form (Section 5.2)	Zone records with crowd count and crowd status
Identify overcrowded zones	Overcrowded Zones and Near Capacity Zones reports (Section 5.3)	Filtered reports display zones by crowd status
Generate reports for organisers	Ten filtered reports (Section 5.3)	Reports covering events, attendees, movements and zones
Present information through dashboard	Dashboard page (Section 5.4)	Single-page summary of event operations

8.3 Comparison with the Existing System

Table 8.2: Existing System versus Proposed Application

Parameter	Existing System	Proposed Application	Improvement
Data Storage	Spreadsheets and paper-based records	Centralised Zoho Creator built-in database	Single source of truth for all event data
Attendee Registration	Printed forms or manual spreadsheet entry	Structured form with field validation and lookup	Reduced errors; direct link to event record
Entry/Exit Tracking	Handwritten gate logs	Digital Check-In form with gate and time fields	Searchable, filterable movement records
Crowd Monitoring	Informal verbal estimates	Zone Monitoring form with numeric crowd count	Structured, comparable zone data
Report Generation	Manual consolidation from multiple files	Built-in filtered reports generated automatically	Instant access to filtered views
Access Control	None; all staff access same files	Role-based access control via Zoho Creator profiles	Each role accesses only permitted functions
Accessibility	Local files only; not shared in real time	Cloud-based; accessible from any browser	Multi-user concurrent access from venue

8.4 Limitations of the Low-Code Approach

The primary limitation of developing this application using Zoho Creator is that user interface customisation is restricted to the options provided by the platform's form builder and page editor. Custom layouts and advanced styling require the use of the Zoho Creator HTML component or integration with Zoho Sites, which was outside the scope of this project. Additionally, the application does not support real-time data updates without manual page refresh, as no live data synchronisation was implemented. Report chart types are limited to those available in Zoho Creator's built-in report editor.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

The Smart Event Check-In and Crowd Flow Management System was built entirely using Zoho Creator's visual configuration tools, placing it in the no-code to low-code range of the development spectrum. Form design, lookup configuration, report creation and role permission setup were all accomplished without writing any code, demonstrating the no-code capability of the platform. Future enhancements involving Deluge scripting for automated crowd status classification or duplicate entry prevention would move the application into the low-code range. Integration with external APIs or custom business logic beyond the platform's built-in features would require pro-code development using Zoho Catalyst or an external backend service.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The Smart Event Check-In and Crowd Flow Management System successfully demonstrates the application of low-code development using Zoho Creator to address a practical event management challenge. The application centralises event information, attendee registrations, check-in and check-out records, and zone-level crowd monitoring data in a single platform, eliminating the need for separate spreadsheets and paper-based records. Four structured forms, ten filtered reports and a dashboard provide event organisers with the information required to manage event operations and identify crowd conditions across different zones.

The application was developed using Zoho Creator's visual form builder, lookup field configuration, report generator and role-based access control, without the use of Deluge scripting or external integrations. Functional testing was conducted against twelve test scenarios covering all core features of the application. The project demonstrates the capability of a low-code platform to deliver a functional, multi-user business application within a short development timeline and without specialised programming knowledge.

9.2 Future Enhancements

- QR code-based attendee verification at check-in gates, enabling automatic validation of registration status without manual lookup.
- Automatic entry and exit tracking using QR code scanners integrated with the Check-In and Check-Out form via Zoho Creator webhooks or Deluge integration scripts.
- CCTV-based crowd detection using computer vision to automatically estimate crowd counts in each zone and update Zone Monitoring records.

9.3 Learning Outcomes and Career Relevance

This project provided practical experience in low-code application development using Zoho Creator. Key skills developed include data modelling using forms and lookup relationships, report design and filtering, dashboard configuration and role-based access control implementation. These skills are directly applicable to roles such as citizen developer, low-code application developer and business analyst in organisations that use Zoho Creator or similar platforms (Microsoft Power Apps, Salesforce Lightning App Builder, AppSheet). The project also reinforced the importance of requirements analysis, structured testing and documentation in the software development process, skills that are transferable to any software development career pathway.

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APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
I	Application created & sections/menu organised	Y	dashboard	5.1	[ ]
II	Forms with ≥ 8 distinct field types	Y	Events, Attendees, Check-In Logs, Zone Monitoring forms	5.2	[ ]
II	Lookup field (relationship between forms)	Y	Events, Attendees, Check-In Logs, Zone Monitoring forms	4.2, 5.2	[ ]
II	Subform (parent–child data)	N	-	4.2, 5.2	[ ]
II	Field validations / layout rules	Y	Required field validation on all forms	5.2	[ ]
II	List report with filter, grouping, custom action	Y	All ten reports in the Reports module	5.3	[ ]
II	Kanban / Calendar / Timeline / Map report	N	-	5.3	[ ]
II	Pivot table / chart report	N	-	5.3	[ ]
II	Page / dashboard built with UI Builder	Y	Dashboard page (Section 5.4)	5.4	[ ]
II	Deluge – form events (validate / success / field actions)	Y	Events, Attendees, Check-In Logs, Zone Monitoring forms	5.5	[ ]
II	Deluge – functions, fetch, loops, collections	N	Not implemented – Not Applicable	5.5	[ ]
II	Email / in-app notifications	N	-	5.5, 5.6	[ ]
II	Form workflow	N	Not implemented – Not Applicable	5.6	[ ]
II	Scheduled workflow	N	Not implemented – Not Applicable	5.6	[ ]
II	Approval process	N	-	5.6	[ ]

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
II	Blueprint	N	-	5.6	[ ]
II	Batch / report workflow	N	Not implemented – Not Applicable	5.6	[ ]
II	Testing & debugging evidence	N	-	6	[ ]
III	Integration with a Zoho service	N	Not implemented – Not Applicable	5.7	[ ]
III	Connection / REST API call (invokeurl)	N	Not implemented – Not Applicable	5.7	[ ]
III	Webhook or Creator REST API	N	Not implemented – Not Applicable	5.7	[ ]
III	Users, roles and permission profiles	Y	Four roles configured in Users and Roles section	5.8	[ ]
III	Record-level access criteria / customer portal	N	-	5.8	[ ]
III	Zia / AI feature	N	-	5.9	[ ]
III	Analytics (charts / Zoho Analytics)	N	-	5.9	[ ]
IV	Performance optimisation applied & measured	N	-	7.2	[ ]
IV	Deployment across environments	N	-	7.1	[ ]
IV	Security controls (encryption, OAuth, permissions)	Y	Four roles configured in Users and Roles section	7.3	[ ]
IV	Governance (audit trail, administration)	N	-	7.4	[ ]
V	Pro-code extension (Catalyst / widgets / custom functions)	N	-	5.10	[ ]
V	Low-code limitations & spectrum analysis	N	-	8.4, 8.5	[ ]
—	Mobile app / portal access	N	-	5.11	[ ]

Total features implemented: 28 / 32

APPENDIX B – COURSE OUTCOME MAPPING

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Ch. 1, 2	[ ]	
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5 (5.1–5.6), 6	[ ]	
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Ch. 4.6, 5.7–5.9	[ ]	
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7	[ ]	
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Ch. 5.9, 5.10, 8.4, 8.5, 9.3	[ ]	

APPENDIX C – COMPLETE DELUGE SOURCE CODE

C.1 Deluge Script DS-01: Auto Evaluate Crowd Status (Zone Monitoring Form)
Deluge Script DS-02: Overcrowding Alert Notification (Zone Monitoring Form)
Deluge Script DS-03: Duplicate Check-In Prevention (Check-In and Check-Out Form)

```
// Script ID : [DS-01]
// Location : [Form → Event / Function name]
// Purpose : [ ]

[paste code here]
```

C.2 Deluge Script DS-01: Auto Evaluate Crowd Status (Zone Monitoring Form)

Deluge Script DS-02: Overcrowding Alert Notification (Zone Monitoring Form)

Deluge Script DS-03: Duplicate Check-In Prevention (Check-In and Check-Out Form)

APPENDIX D – USER MANUAL

D.1 Accessing the Application

1. Open a web browser and navigate to the Zoho Creator application URL: [you fill this].
2. Sign in using your Zoho account credentials.
3. The application home page will be displayed after successful login.
4. For mobile access, open the same URL in a mobile browser. Zoho Creator renders a responsive interface compatible with mobile screens.

D.2 Instructions for Event Organiser

- 1. Open a supported web browser and access the live application URL: [you fill this].
- 2. Log in using your assigned Event Organiser credentials provided by the system administrator.

D.3 Instructions for Check-In Staff

- 1. Open a mobile browser or workstation at your assigned entry gate and log in at: [you fill this].
- 2. Sign in using your Check-In Staff credentials.

3. Open the Check-In and Check-Out form.

APPENDIX E – EVALUATION RUBRIC

(For Examiner Use Only)

Student Name	Hoshni Sivaprakash	Register No.	241401029
Project Title	Smart Event Check-In & Crowd Flow Management	Date	

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	
	TOTAL		**100**	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner

Name: _____

Signature of External Examiner

Name: _____

APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	Hoshni Sivapraksh	Register No.	241401029
Department / Section	Computer Science and business Systems	Date of Viva	
Application URL Verified	Yes / No	Demo Duration	_____min

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	**100**	

Examiners' Remarks:

_____

_____

INTERNAL EXAMINER **EXTERNAL EXAMINER**

Name: _____ Name: _____